

EFFECTIVE ADAPTIVE BOUNDARY IDENTIFICATION OF INFINITE DAMPED JACOBI LATTICES

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ABSTRACT. We study identification of a damped half-line lattice from a single noisy boundary coordinate under chronological feedback. For a homogeneous bath, six short response values identify five physical coefficients. Finite-prefix calibration and fresh signs give uniform contrast without resetting the state, leading to a random-information quasi-Bernstein-von Mises theorem, strong filter jets, and a functional posterior for the memory kernel. For a fully unknown Jacobi stiffness sequence, we derive a constructive response-jet inverse with condition bound $Q^{64(J+1)^2}$ through depth J , using a finite moment recursion and explicitly bounded Gram solves. A multiscale sampling law then yields response separation at least $\exp\{-A_0(J+1)^3\}\delta^{20(J+1)}$. Policy-uniform finite-sample likelihood bounds give shrinking coefficient recovery for $J_n = o((\log n)^{1/3})$ and contraction in an exponentially weighted operator norm. The washed experiment has a full posterior LDP; a no-washout experiment supports arbitrary bounded adaptive baselines in linear physical time. In the latter, a posterior Laplace-tracking theorem identifies full LDPs along convergent conditional-information subsequences, while preserving an unconditional information-floor upper bound. Simultaneous frequentist confidence cylinders are valid at all observation times by a stopped exponential-supermartingale argument.

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Date: September 5, 2026.

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1. RESPONSE GEOMETRY AND THE TWO EXPERIMENTAL CLOCKS

The unknown coefficients of a damped lattice enter the boundary response through repeated propagation into its unobserved bath. Deterministic inverse geometry and chronological statistical information must therefore be proved separately and then quantitatively connected. This article makes that connection for a homogeneous five-parameter bath and for a complete inhomogeneous Jacobi coefficient sequence.

In the homogeneous model, the response jets form an explicitly invertible coordinate system for boundary damping and pinning, coupling strength, and bath damping and pinning. A six-duration interpolation converts the jets into a global response embedding. Calibration occupies a positive fraction of every finite prefix, with the duration fixed before a fresh random sign. This chronology prevents an existing hidden state from cancelling the conditional information. Uniform smooth Gaussian likelihood estimates give a total-variation quasi-Bernstein–von Mises theorem without requiring a limit for the realized information matrix. The initially unknown state is integrated against an arbitrary probability on a fixed Hilbert ball; its likelihood contribution has a summable derivative budget. The same estimates control current-state filter jets in fixed separable dual spaces and push the finite-dimensional posterior through the entire memory-kernel map.

In the inhomogeneous model every positive off-diagonal coefficient and every pinning coefficient is unknown, as is one common damping parameter. The parameter space is a compact product, not a finite-dimensional truncation. Classical Weyl–Schur reconstruction supplies identifiability. For quantitative inversion we use the exact response-to-moment generating identity, the coefficient norm of monic orthogonal polynomials, and a ledger of each finite Gram solve, quadratic form, ratio, and square root. The resulting bound is

$$d_J(\beta, \gamma) \leq Q^{64(J+1)^2} \max_{r \leq 4J+8} |h_\beta^{(r)}(0) - h_\gamma^{(r)}(0)|.$$

A finite Taylor grid and an explicitly weighted multiscale design give

$$d_J(\beta, \gamma) \geq \delta \implies D(\beta, \gamma) \geq e^{-A_0(J+1)^3} \delta^{20(J+1)} \quad (0 < \delta \leq 1).$$

Both constants are uniform in the infinite tail, and the proof includes the shallow depths. The rates are sufficient rates; no optimality is asserted.

The central probability estimate is finite-sample and policy-uniform. With $\mathcal{H}_d$ the entropy of a deterministic metric controlling every candidate mean field, it bounds the score and centered conditional-square errors by

$$4 \exp\{\mathcal{H}_d(e(t)) - nt^2/(8V)\} + e^{-n}.$$

Together with an explicit integrated-likelihood transient tail and a finite-cylinder prior-mass bound, this yields a posterior separation inequality at shrinking t . It proves coefficient recovery at the concrete radius $\delta_n = \exp\{-\eta \log n/[80(J_n + 1)]\}$ whenever $(J_n + 1)^3 = o(\log n)$, for any fixed $0 < \eta < 1$.

The article distinguishes two clocks. Logarithmic zero-force washouts give summable residuals, a full good posterior LDP with sampled-response rate, and total physical time $n \log n$. A complementary no-washout protocol uses a predictable baseline followed by a fresh signed pulse. It has linear physical time and conditional information at least the same exploration floor. Its actual

information can be larger and need not converge. We therefore prove a Laplace approximation following the entire conditional information sequence, with full LDPs along each uniform cluster subsequence; a full-sequence adaptive LDP follows when that sequence converges. The universal closed-set information-floor bound requires no such convergence. Finally, stopped exponential supermartingales give confidence sets simultaneous over all times and countably many diagnostic cells. These are frequentist confidence sets, not claims about Bayesian credible-set coverage.

1.1. Relation to established theory. Inverse spectral reconstruction, the Jacobi moment problem, and orthogonal polynomial recurrences are classical; Teschl [1] is a standard reference. We use these identities as a deterministic engine and provide the finite-map condition calculation needed for the present statistical bounds. The general principle of transferring forward regression consistency through inverse stability also predates this work; see Vollmer [2]. Here the forward data arise from one chronological boundary trajectory, and a common metric controls all bounded feedback convolutions.

Posterior asymptotics for dependent data and misspecified models, including posterior large-deviation behavior, have a broader existing theory, notably Shalizi [3]. Our compact-space Laplace step is elementary once the uniform likelihood estimate is established. The detailed work here is the explicit shrinking-net bound, the uniform transient correction, and their mechanical verification. Time-uniform confidence sequences and exponential-supermartingale methods likewise have an established literature; see Howard, Ramdas, McAuliffe and Sekhon [4]. We use a countable visit-time allocation to connect that mechanism to the same finite-grid inverse, rather than interpreting a credible set as a confidence set. The finite-dimensional Gaussian theorem is a separate regular likelihood result; its state and memory images are finite-rank Gaussian images, not an infinite-dimensional Bernstein–von Mises claim.

2. A UNIFORM NONLINEAR ADAPTIVE GAUSSIAN THEOREM

This section defines the triangular experiment once and proves the probability and Laplace estimates used later. Its role is not to replace model-specific identification. The mechanical verification of the global contrast is given in Proposition 3.4.

2.1. Experiment class and two-stage filtration. Let $\Theta \subset \mathbb{R}^p$ be compact and convex, and let $\Theta_\circ \Subset \text{int } \Theta$. The hidden state space $\mathbf{H}$ is a separable Hilbert space. Fix $R < \infty$, noise variance $\sigma^2 > 0$, and an integer $s > p/2$.

Definition 2.1 (Triangular adaptive experiment). For every terminal size n , an admissible policy κ_n constructs random variables in the chronological order

$$\mathcal{F}_{i-1} \longrightarrow A_i \longrightarrow \xi_i \longrightarrow Y_i, \quad i = 1, \dots, n.$$

Here A_i is the action, possibly selected with a fresh policy randomizer,

$$\mathcal{G}_i = \mathcal{F}_{i-1} \vee \sigma(A_i), \quad \mathcal{F}_i = \mathcal{G}_i \vee \sigma(Y_i),$$

and $\xi_i \sim N(0, \sigma^2)$ is independent of $\mathcal{G}_i$. For every $\theta \in \Theta$, the zero-initial-state mean $m_{n,i}(\theta)$ is a $\mathcal{G}_i$ -measurable random C^{s+3} function of θ . The transient contribution of initial state z is $b_{n,i}(\theta, z)$, also $\mathcal{G}_i$ -measurable. The external true law is

$$Y_i = m_{n,i}(\theta_0) + b_{n,i}(\theta_0, z_0) + \xi_i, \quad \theta_0 \in \Theta_\circ, \quad \|z_0\| \leq R. \quad (1)$$

The policy kernel is the same measurable function of the observed record under every parameter; it cannot use θ_0 or z_0 as an oracle. It may depend on n .

For a working initial-state probability ν_n , supported on the radius R ball but otherwise arbitrary, define

$$L_n^{\nu_n}(\theta) = \int_{\mathbf{H}} \prod_{i=1}^n \varphi_{\sigma}\{Y_i - m_{n,i}(\theta) - b_{n,i}(\theta, z)\} \nu_n(dz), \quad (2)$$

$$\Pi_n^{\nu_n}(d\theta) \propto \pi(\theta) L_n^{\nu_n}(\theta) d\theta. \quad (3)$$

Policy factors cancel in the chronological joint likelihood because the same kernel is evaluated at the realized past. If $z_0 \notin \text{supp } \nu_n$, (3) is a quasi-posterior under the external truth (1). No density on $\mathbf{H}$ is assumed.

The experiment index is

$$E = (n, \kappa_n, \theta_0, z_0, \nu_n) \in \mathfrak{E}_n.$$

All random variables below are understood under $P_E = P_{\theta_0, z_0}^{\kappa_n}$. The following notation makes every uniform quantifier explicit.

Definition 2.2 (Uniform stochastic orders). For random variables Z_E , write $Z_E = o_{\mathfrak{E}}(1)$ when, for every $\eta > 0$,

$$\sup_{E \in \mathfrak{E}_n} P_E(|Z_E| > \eta) \longrightarrow 0.$$

Write $Z_E = O_{\mathfrak{E}}(1)$ when

$$\lim_{K \rightarrow \infty} \sup_{n \geq 1} \sup_{E \in \mathfrak{E}_n} P_E(|Z_E| > K) = 0.$$

The same definitions apply to random norms and suprema of random fields.

Assumption 2.3 (Uniform smoothness, transience, and contrast). *There are deterministic constants $B < \infty$, $c_0 > 0$, a summable sequence (d_i) , and $\varepsilon_n \downarrow 0$, independent of the experiment index, such that:*

(1) for $|\alpha| \leq s + 3$,

$$\sup_{n, i, E, \theta} |\partial_{\theta}^{\alpha} m_{n,i}(\theta)| \leq B;$$

(2) for $|\alpha| \leq s + 3$,

$$\sup_{n, E, \theta, \|z\| \leq R} |\partial_{\theta}^{\alpha} b_{n,i}(\theta, z)| \leq d_i, \quad \sum_{i \geq 1} d_i < \infty;$$

(3) with probability at least $1 - \varepsilon_n$, simultaneously in $\theta \in \Theta$,

$$\frac{1}{n} \sum_{i=1}^n \{m_{n,i}(\theta) - m_{n,i}(\theta_0)\}^2 \geq c_0 |\theta - \theta_0|^2. \quad (4)$$

The prior density π is continuous on Θ , bounded above, and bounded below on a fixed neighbourhood of Θ_0 .

The summability hypothesis is stated with one deterministic envelope. It implies $|Y_i| \leq B_0 + |\xi_i|$ for every i in the applications below; equivalently this bound may be included in the experiment definition.

2.2. The exact nuisance factor and misspecification. Set

$$L_n^0(\theta) = \prod_{i=1}^n \varphi_{\sigma}\{Y_i - m_{n,i}(\theta)\}.$$

Completing each Gaussian square gives the exact identity

$$\begin{aligned} L_n^{\nu_n}(\theta) &= L_n^0(\theta) e^{R_n^{\nu_n}(\theta)}, \\ R_n^{\nu_n}(\theta) &= \log \int e^{V_n^{\theta}(z)} \nu_n(dz), \end{aligned} \quad (5)$$

$$V_n^\theta(z) = \sigma^{-2} \sum_{i=1}^n \left[(Y_i - m_{n,i}(\theta)) b_{n,i}(\theta, z) - \frac{1}{2} b_{n,i}(\theta, z)^2 \right]. \quad (6)$$

Lemma 2.4 (Uniform nuisance derivative budget). *For every fixed $r \leq s + 3$ and $q < \infty$, there is a random variable $W_{n,r,E}$ satisfying*

$$\sup_{n,E} \mathbb{E}_E W_{n,r,E}^q < \infty$$

and

$$\sup_{\theta, \|z\| \leq R} \sum_{|\alpha| \leq r} \left| \partial_\theta^\alpha V_n^\theta(z) \right| \leq W_{n,r,E}, \quad \|R_n^{\nu_n}\|_{C^r(\Theta)} \leq C_r (1 + W_{n,r,E})^{r+1}. \quad (7)$$

The constants are independent of ν_n .

Proof. Every differentiated summand in (6) is bounded by $C_r d_i (1 + |\xi_i|)$. Thus one may take

$$W_{n,r,E} = C_r \sum_{i \geq 1} d_i (1 + |\xi_i|).$$

Minkowski's inequality and the Gaussian moments prove the uniform moment bound. Derivatives of a log integral are finite sums of cumulants of the bounded derivatives of V_n under the positive normalized tilt $e^{V_n} \nu_n / \int e^{V_n} d\nu_n$. Their absolute values are bounded by a universal polynomial in $1 + W_{n,r,E}$, proving (7) without any regularity assumption on ν_n . $\square$

To state the pseudo-true conclusion, let $\theta_{n,E}^*$ be any minimizer of the expected negative marginal log likelihood, normalized by n .

Proposition 2.5 (The transient pseudo-true parameter). *Suppose the failure probability in (4) is exponentially small. Then*

$$\sup_{E \in \mathfrak{E}_n} \left| \theta_{n,E}^* - \theta_0 \right| \leq C n^{-1}.$$

In particular the asymptotic pseudo-true parameter is the physical parameter θ_0 , even when $z_0 \notin \text{supp } \nu_n$.

Proof. The expected zero-state log-likelihood contrast is

$$\frac{1}{2\sigma^2 n} \mathbb{E}_E \sum_i \{m_{n,i}(\theta) - m_{n,i}(\theta_0)\}^2 - \frac{1}{\sigma^2 n} \mathbb{E}_E \sum_i b_{n,i}(\theta_0, z_0) \{m_{n,i}(\theta) - m_{n,i}(\theta_0)\}.$$

The first term is at least $c|\theta - \theta_0|^2$; the exponentially small complement is harmless because all means are bounded. The second is bounded by $C|\theta - \theta_0|/n$. By Lemma 2.4, the expected difference $R_n^{\nu_n}(\theta) - R_n^{\nu_n}(\theta_0)$, divided by n , has the same linear bound. A minimizer therefore obeys $cr^2 \leq Cr/n$, which proves the assertion. $\square$

2.3. Smooth martingale fields. For a multi-index β , put

$$S_{n,\beta}(\theta) = \sum_{i=1}^n \xi_i \partial_\theta^\beta m_{n,i}(\theta).$$

Lemma 2.6 (Policy-uniform Sobolev score bound). *For every $|\beta| \leq 3$,*

$$\mathbb{E}_E \|S_{n,\beta}\|_{C(\Theta)}^2 \leq Cn \quad \text{uniformly in } E \in \mathfrak{E}_n. \quad (8)$$

Consequently $n^{-1/2} \|S_{n,\beta}\|_{C(\Theta)} = O_{\mathfrak{E}}(1)$.

Proof. Use the fixed bounded convex domain $\text{int } \Theta$, which has the cone property, and its deterministic Sobolev embedding constant. For each $|\alpha| \leq s$, martingale orthogonality from $\mathbb{E}(\xi_i | \mathcal{G}_i) = 0$ gives

$$\mathbb{E}_E \int_\Theta |\partial_\theta^\alpha S_{n,\beta}(\theta)|^2 d\theta = \sigma^2 \sum_{i=1}^n \mathbb{E}_E \int_\Theta \left| \partial_\theta^{\alpha+\beta} m_{n,i}(\theta) \right|^2 d\theta \leq Cn.$$

The deterministic Sobolev embedding $H^s \hookrightarrow C$, valid because $s > p/2$, proves (8). $\square$

Write

$$g_i = Dm_{n,i}(\theta_0), \quad I_n = \frac{1}{n\sigma^2} \sum_{i=1}^n g_i g_i^T, \quad \Delta_n = \frac{1}{\sigma^2 \sqrt{n}} \sum_{i=1}^n g_i \xi_i.$$

The contrast event implies, by division along a line through θ_0 ,

$$I_n \geq (c_0/\sigma^2)\text{Id}.$$

The smoothness bound gives $I_n \leq (B^2 p/\sigma^2)\text{Id}$, and Lemma 2.6 gives $\Delta_n = O_{\mathfrak{E}}(1)$.

2.4. Local quadratic expansion and global localization.

Proposition 2.7 (Uniform local expansion and global envelope). *For every fixed M ,*

$$\sup_{|h| \leq M} \left| \log \frac{L_n^{\nu_n}(\theta_0 + h/\sqrt{n})}{L_n^{\nu_n}(\theta_0)} - h^T \Delta_n + \frac{1}{2} h^T I_n h \right| = o_{\mathfrak{E}}(1). \quad (9)$$

On the contrast event there is $Z_{n,E} = O_{\mathfrak{E}}(1)$ such that, simultaneously in $\theta \in \Theta$,

$$\log \frac{L_n^{\nu_n}(\theta)}{L_n^{\nu_n}(\theta_0)} \leq -c_1 n |\theta - \theta_0|^2 + Z_{n,E} \sqrt{n} |\theta - \theta_0|, \quad c_1 = \frac{c_0}{2\sigma^2}. \quad (10)$$

Consequently, for every $\eta > 0$,

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{E \in \mathfrak{E}_n} P_E \{ \Pi_n^{\nu_n}(\sqrt{n} |\theta - \theta_0| > M) > \eta \} = 0. \quad (11)$$

Proof. Put $d_i(\theta) = m_{n,i}(\theta) - m_{n,i}(\theta_0)$. The exact zero-state log-likelihood ratio under the external truth is

$$\sigma^{-2} \sum_i \xi_i d_i(\theta) + \sigma^{-2} \sum_i b_{n,i}(\theta_0, z_0) d_i(\theta) - \frac{1}{2\sigma^2} \sum_i d_i(\theta)^2. \quad (12)$$

The fundamental theorem of calculus and Lemma 2.6 bound its innovation term by $O_{\mathfrak{E}}(\sqrt{n}) |\theta - \theta_0|$, uniformly in θ . The true transient cross term is at most $C(\sum_{i \geq 1} d_i) |\theta - \theta_0|$, where the d_i 's inside the summation are the deterministic transient envelope from Assumption 2.3, not the mean increment $d_i(\theta)$. Lemma 2.4 bounds the difference of the two R_n terms by $O_{\mathfrak{E}}(1) |\theta - \theta_0|$. Combining with (4) proves (10).

For $\delta = h/\sqrt{n}$, Taylor expansion gives

$$d_i(\theta_0 + \delta) = g_i^T \delta + \frac{1}{2} \delta^T D^2 m_{n,i}(\theta_0) \delta + O(|\delta|^3).$$

The innovation multiplied by the Hessian is $O_{\mathfrak{E}}(n^{-1/2})$ by the score lemma, its third-order remainder is bounded by $Cn^{-3/2} \sum_i |\xi_i| = O_{\mathfrak{E}}(n^{-1/2})$, and the quadratic-sum remainder is $O(n|\delta|^3) = O(n^{-1/2})$ on a fixed h -ball. The true transient and R_n differences are also $O_{\mathfrak{E}}(n^{-1/2})$. This proves (9).

On events where $Z_{n,E} \leq K$, the rescaled right side of (10) is $-c_1 |h|^2 + K|h|$. The local expansion integrated over $|h| \leq 1$, together with the lower bound on π , gives a denominator at least $n^{-p/2} e^{-O_{\mathfrak{E}}(1)}$. The Gaussian integral of the global envelope outside $|h| > M$ tends to zero uniformly as $M \rightarrow \infty$. Tightness and the vanishing contrast-failure probability prove (11). $\square$

2.5. A relative random-information Laplace lemma.

Lemma 2.8 (Uniform relative Laplace principle). *Let $K_{n,E} \geq 0$ on $H_{n,E} = \sqrt{n}(\Theta - \theta_0)$. Suppose, uniformly in the experiment class, that the eigenvalues of $J_{n,E}$ lie in $[\underline{j}, \bar{j}]$ with probability tending to one, $U_{n,E} = O_{\mathfrak{E}}(1)$, and for every fixed M ,*

$$\sup_{|h| \leq M} |\log K_{n,E}(h) - h^T U_{n,E} + \frac{1}{2} h^T J_{n,E} h| = o_{\mathfrak{E}}(1).$$

Suppose also $K_{n,E}(h) \leq C_{n,E}e^{-a|h|^2+B_{n,E}|h|}$, with $C_{n,E}, B_{n,E} = O_{\mathfrak{E}}(1)$, on these events. Let $r_{n,E} \geq 0$ be uniformly bounded and converge uniformly to one on each fixed ball. Extend $K_{n,E}r_{n,E}$ by zero outside $H_{n,E}$, and put $q_{n,E}(h) = e^{h^T U_{n,E} - h^T J_{n,E} h/2}$. Then

$$\frac{\int_{\mathbb{R}^p} |K_{n,E}r_{n,E} - q_{n,E}| dh}{\int_{\mathbb{R}^p} q_{n,E} dh} = o_{\mathfrak{E}}(1), \quad \int q_{n,E} dh = (2\pi)^{p/2} (\det J_{n,E})^{-1/2} e^{U_{n,E}^T J_{n,E}^{-1} U_{n,E}/2}. \quad (13)$$

Proof. Restrict to the event $|U| + B + C \leq K$ and the stated spectral bounds. The Gaussian integral, evaluated by completing the square, is bounded above and below by positive constants depending only on K and the fixed spectral interval. On $|h| \leq M$, exponentiation of the uniform log error and the prior-ratio error gives vanishing L^1 difference. Outside this ball, both the Gaussian and the common quadratic envelope have integrals tending uniformly to zero as $M \rightarrow \infty$. The scaled domains exhaust every fixed ball uniformly because the truth lies in Θ_0 . Divide by the lower Gaussian integral bound, then let $n \rightarrow \infty$, $M \rightarrow \infty$, and $K \rightarrow \infty$. Relative L^1 error controls the normalizers and hence total variation. $\square$

Theorem 2.9 (Uniform nonlinear adaptive quasi-Bernstein–von Mises). *Under Assumption 2.3, uniformly over $E \in \mathfrak{E}_n$,*

$$d_{\text{TV}} \left(\mathcal{L}\{\sqrt{n}(\theta - \theta_0) \mid \mathcal{F}_n\}, N(I_n^{-1} \Delta_n, I_n^{-1}) \right) = o_{\mathfrak{E}}(1). \quad (14)$$

Define the comparison law arbitrarily when I_n is singular, an event of uniformly vanishing probability. The evidence obeys

$$\frac{\mathcal{Z}_n^{\nu_n}}{L_n^{\nu_n}(\theta_0)} = n^{-p/2} \pi(\theta_0) (2\pi)^{p/2} (\det I_n)^{-1/2} e^{\Delta_n^T I_n^{-1} \Delta_n/2} [1 + o_{\mathfrak{E}}(1)]. \quad (15)$$

Neither assertion assumes convergence of I_n .

Proof. Apply Lemma 2.8 to the likelihood ratio at $\theta_0 + h/\sqrt{n}$, with $J = I_n$, $U = \Delta_n$, and $r(h) = \pi(\theta_0 + h/\sqrt{n})/\pi(\theta_0)$. Proposition 2.7 supplies the local expansion and Gaussian envelope; contrast and the score lemma supply the spectral and tightness hypotheses. Prior continuity and positivity give the ratio condition. Normalize the relative L^1 bound to obtain total variation, and retain its integral with the Jacobian $n^{-p/2}$ to obtain evidence. $\square$

3. FIVE PHYSICAL COEFFICIENTS FROM ONE BOUNDARY COORDINATE

Let $\mathcal{H} = \ell^2(\mathbb{N}_0; \mathbb{R})$, $\mathbf{H} = \mathcal{H} \oplus \mathcal{H}$, $Pq = \langle q, e_0 \rangle e_0$, and $Q_0 = I - P$. The half-line Laplacian is $(Lq)_0 = q_0 - q_1$ and $(Lq)_j = 2q_j - q_{j-1} - q_{j+1}$ for $j \geq 1$; it is nonnegative, self-adjoint, and bounded by four. Let all endpoints of the fixed box

$$\vartheta = (c, k, \epsilon, c_b, k_b) \in \Theta = [c_-, c_+] \times [k_-, k_+] \times [\epsilon_-, \epsilon_+] \times [c_{b,-}, c_{b,+}] \times [k_{b,-}, k_{b,+}]$$

be finite and strictly positive, with nonempty interior. Set

$$C_{\vartheta} = cP + c_b Q_0, \quad K_{\vartheta} = kP + k_b Q_0 + \epsilon L, \quad A_{\vartheta} = \begin{pmatrix} 0 & I \\ -K_{\vartheta} & -C_{\vartheta} \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ e_0 \end{pmatrix}, \quad \ell(q, v) = q_0.$$

Thus $\dot{q} = v$, $\dot{v} = -K_{\vartheta}q - C_{\vartheta}v + e_0 u(t)$.

Lemma 3.1 (Uniform physical stability). *There are box constants $M \geq 1$, $\lambda > 0$ such that $\|e^{tA_{\vartheta}}\| \leq M e^{-\lambda t}$ for all $t \geq 0$. Every fixed-order parameter derivative is bounded by $M_r e^{-\lambda_r t}$, for $M_r < \infty$, $\lambda_r > 0$. Every fixed-order derivative of a uniformly bounded-input trajectory is uniformly bounded, independently of its switching history.*

Proof. Let c_* and k_* be the minimum damping and pinning endpoints, and c^* , k^* upper bounds on $C_{\vartheta}, K_{\vartheta}$. Take $d = \min(c_*/2, \sqrt{k_*/2})$ and use the energy

$$E_{\vartheta}(q, v) = \frac{1}{2} \|v\|^2 + \frac{1}{2} \langle K_{\vartheta} q, q \rangle + d \langle q, v \rangle + \frac{d}{2} \langle C_{\vartheta} q, q \rangle.$$

Young's inequality makes this uniformly equivalent to $\|(q, v)\|^2$. Differentiation along the unforced motion gives

$$E'_\vartheta = -\langle (C_\vartheta - dI)v, v \rangle - d\langle K_\vartheta q, q \rangle \leq -\min(c_* - d, dk_*)\|(q, v)\|^2.$$

This proves common exponential stability. The generator is affine in the five parameters with bounded derivatives. Ordered Duhamel integrals for order r are bounded by $C_r t^r e^{-\lambda t}$; decreasing the decay rate absorbs the polynomial. Integrating these bounds against a bounded input proves the final assertion. $\square$

3.1. The response jet and six-duration embedding. Write $h_\vartheta(t) = \ell \int_0^t e^{sA_\vartheta} B ds = \sum_{j \geq 0} r_j t^{j+1}/(j+1)!$, and put $s = k + \epsilon$. Successive application of the banded generator gives

$$\begin{aligned} r_0 &= 0, & r_1 &= 1, & r_2 &= -c, \\ r_3 &= c^2 - s, & r_4 &= c(-c^2 + 2s), \\ r_5 &= c^4 - 3c^2 s + s^2 + \epsilon^2, \\ r_6 &= -c^5 + 4c^3 s - 2c\epsilon^2 - 3cs^2 - c_b \epsilon^2, \\ r_7 &= c^6 - 5c^4 s + 3c^2 \epsilon^2 + 6c^2 s^2 + 2cc_b \epsilon^2 \\ &\quad + c_b^2 \epsilon^2 - 2\epsilon^3 - 2\epsilon^2 s - s^3 - \epsilon^2 k_b. \end{aligned} \tag{16}$$

The map $\mathcal{J}(\vartheta) = (r_2, r_3, r_5, r_6, r_7)$ has the following triangular inverse, for $x = (x_2, x_3, x_5, x_6, x_7)$:

$$c = -x_2, \quad s = c^2 - x_3, \quad \epsilon = (x_5 - c^4 + 3c^2 s - s^2)^{1/2}, \quad k = s - \epsilon, \tag{17}$$

$$c_b = (-c^5 + 4c^3 s - 2c\epsilon^2 - 3cs^2 - x_6)/\epsilon^2, \tag{18}$$

$$\begin{aligned} k_b &= \epsilon^{-2}[c^6 - 5c^4 s + 3c^2 \epsilon^2 + 6c^2 s^2 + 2cc_b \epsilon^2 + c_b^2 \epsilon^2 \\ &\quad - 2\epsilon^3 - 2\epsilon^2 s - s^3 - x_7]. \end{aligned} \tag{19}$$

The radicand is at least ϵ_-^2 on the image. These formulas are therefore smooth on a neighbourhood of the compact image. A smooth cutoff supported in that neighbourhood extends the inverse to a globally bounded smooth map $G : \mathbb{R}^5 \rightarrow \mathbb{R}^5$.

At $t_j = j\tau$, $1 \leq j \leq 6$, remove the known term $t_j^2/2$ and let $y_j = h_\vartheta(t_j) - t_j^2/2$. Define

$$V_{j,\ell-1} = j^{\ell+1}/(\ell+1)!, \quad 2 \leq \ell \leq 7, \quad D_\tau = \text{diag}(\tau^3, \dots, \tau^8), \quad \widehat{R}_\tau = D_\tau^{-1} V^{-1} (y_j)_{j=1}^6.$$

Dividing each row of V by j^3 reduces its determinant to a nonzero Vandermonde determinant, with nonzero column factors. The omitted Taylor series starts at order nine, uniformly also for one parameter derivative. Multiplication by D_τ^{-1} amplifies at most by τ^{-8} . Hence

$$\|\widehat{R}_\tau - (r_2, \dots, r_7)\|_{C^1(\Theta)} \leq C\tau, \quad 0 < \tau \leq 1. \tag{20}$$

Theorem 3.2 (Global six-duration mechanical embedding). *For sufficiently small fixed $\tau > 0$, the response map $H_\tau(\vartheta) = (h_\vartheta(j\tau))_{j=1}^6$ satisfies*

$$|H_\tau(\vartheta) - H_\tau(\vartheta')| \geq b_\tau |\vartheta - \vartheta'|, \quad |DH_\tau(\vartheta)v| \geq b_\tau |v|, \quad b_\tau > 0. \tag{21}$$

Proof. Let P_5 select jet coordinates $(r_2, r_3, r_5, r_6, r_7)$. The smooth reconstructed map $F_\tau = G(P_5 \widehat{R}_\tau)$ obeys $\|F_\tau - \text{Id}\|_{C^1} \leq C\tau$. Choose τ so this norm is at most one half. Integration along a segment in the convex parameter box gives $|F_\tau(\vartheta) - F_\tau(\vartheta')| \geq |\vartheta - \vartheta'|/2$, and the same lower bound holds for tangent vectors. The map taking six responses to F_τ has a finite global Lipschitz constant, because G does. Divide by twice that constant. $\square$

3.2. Finite-prefix calibration and exact chronological contrast. Fix a pulse amplitude $a > 0$, an exploitation-force bound $U < \infty$, exploitation durations in $[\tau_{\min}, \tau_{\max}]$ with $0 < \tau_{\min} \leq \tau_{\max} < \infty$, and constants $\rho \in (0, 1]$, $C_{\text{cal}} \geq 0$. Calibration membership is predictable from $\mathcal{F}_{i-1}$, and

$$N_{\text{cal}}(m) \geq \rho m - C_{\text{cal}}, \quad 1 \leq m \leq n. \quad (22)$$

These constants are common to all horizons and all admissible policies. Calibration slots in their own chronological order are grouped into windows of six. Before drawing the fresh sign, choose the duration predictably from those not yet used in the current window; thus each complete window uses $\tau, 2\tau, \dots, 6\tau$ once. Then apply aS_i and finally observe with independent Gaussian noise. In symbols,

$$\mathcal{F}_{i-1} \longrightarrow s_i \longrightarrow S_i \longrightarrow \xi_i \longrightarrow Y_i.$$

Exploitation may be arbitrary bounded history-dependent control whose full current action is selected before the readout. The class is nonempty even when $C_{\text{cal}} = 0$: a deterministic schedule with $N_{\text{cal}}(m) = \lceil \rho m \rceil$ satisfies the condition.

Lemma 3.3 (Complete windows in every prefix). *Pathwise,*

$$W_n = \lfloor N_{\text{cal}}(n)/6 \rfloor \geq \rho n/6 - C_{\text{win}}, \quad C_{\text{win}} = C_{\text{cal}}/6 + 1. \quad (23)$$

Proof. Apply $\lfloor x \rfloor \geq x - 1$ and (22). Only the last incomplete window is lost. $\square$

For zero-state means under the entire realized action history, at a calibration slot

$$m_i^\vartheta - m_i^{\vartheta_0} = D_{i,s_i}(\vartheta, \vartheta_0) + aS_i\{h_\vartheta(s_i) - h_{\vartheta_0}(s_i)\}, \quad (24)$$

where D_{i,s_i} is $\mathcal{F}_{i-1}$ -measurable. Conditional expectation of its square is

$$D_{i,s_i}^2 + a^2\{h_\vartheta(s_i) - h_{\vartheta_0}(s_i)\}^2. \quad (25)$$

Summing over a complete window gives at least $a^2b_\tau^2|\vartheta - \vartheta_0|^2$, regardless of the adaptive order of its durations.

Proposition 3.4 (Full-index empirical contrast). *Let $\bar{c}_\tau = \rho a^2 b_\tau^2/6$. For every fixed $0 < c_* < \bar{c}_\tau$, common constants $C_1, C_2 > 0$ satisfy*

$$P_E \left\{ \inf_{\vartheta \neq \vartheta_0} \frac{n^{-1} \sum_i (m_i^\vartheta - m_i^{\vartheta_0})^2}{|\vartheta - \vartheta_0|^2} < c_* \right\} \leq C_1 e^{-C_2 n}. \quad (26)$$

On the same good event, $n^{-1} \sum_i (v^T D m_i^{\vartheta_0})^2 \geq c_ |v|^2$ for all v .*

Proof. Use the compact radial index set

$$\mathcal{K} = \{(\theta, r, v) : \theta \in \Theta, v \in S^4, r \geq 0, \theta + rv \in \Theta\}, \quad q_i(\theta, r, v) = \int_0^1 v^T D m_i^{\theta + sr v} ds.$$

For $r > 0$, this is the divided mean difference; at zero it is the tangent derivative. Lemma 3.1 bounds q_i and two parameter derivatives uniformly, hence q_i^2 is bounded and Lipschitz on this fixed compact set, independently of the policy and horizon. Let $X_i = q_i^2 - E[q_i^2 | \mathcal{F}_{i-1}]$. The window identity and count give, for every index,

$$\sum_i E[q_i^2 | \mathcal{F}_{i-1}] \geq \bar{c}_\tau n - a^2 b_\tau^2 C_{\text{win}}.$$

Each X_i is a bounded martingale difference with a common Lipschitz constant L_X . A deterministic finite δ -net and Azuma's inequality give, except on probability $C' e^{-c'n}$, $\inf_{\mathcal{K}} n^{-1} \sum_i X_i \geq -\eta - L_X \delta$. Choose η, δ with $\eta + L_X \delta < (\bar{c}_\tau - c_*)/2$. For all sufficiently large n , the displayed predictable lower bound then proves the assertion. Enlarge C_1 for the remaining finite number of horizons. Taking $r = 0$ gives the tangent assertion. $\square$

3.3. The mechanical posterior and evidence. Define

$$x_i^\vartheta = \int_0^{t_i} e^{(t_i-s)A_\vartheta} B u(s) ds, \quad m_i^\vartheta = \ell x_i^\vartheta, \quad b_i^\vartheta(z) = \ell e^{t_i A_\vartheta} z.$$

Every block has length at least $\underline{\tau} = \min(\tau, \tau_{\min}) > 0$, so $t_i \geq i\underline{\tau}$. Every fixed derivative of b_i is bounded by a common summable exponential envelope. Put

$$I_n = (n\sigma^2)^{-1} \sum_i Dm_i^{\vartheta_0} (Dm_i^{\vartheta_0})^T, \quad \Delta_n = (\sigma^2 \sqrt{n})^{-1} \sum_i Dm_i^{\vartheta_0} \xi_i.$$

Theorem 3.5 (Five-parameter mechanical quasi-Bernstein–von Mises). *Let the true parameter range over a fixed compact interior subset of Θ , let $\|z_0\| \leq R$, and let ν_n be arbitrary working probabilities on the same Hilbert ball. Assume the parameter-prior conditions of Assumption 2.3. Uniformly over these choices and all prefix-balanced policies, including horizon-dependent ones,*

$$d_{\text{TV}} \left(\mathcal{L}\{\sqrt{n}(\vartheta - \vartheta_0) \mid \mathcal{F}_n\}, N(I_n^{-1} \Delta_n, I_n^{-1}) \right) = o_{\mathfrak{E}}(1), \quad (27)$$

with the singular-event convention of Theorem 2.9, and

$$\frac{\mathcal{Z}_n^{\nu_n}}{L_n^{\nu_n}(\vartheta_0)} = \frac{(2\pi)^{5/2} \pi(\vartheta_0)}{n^{5/2} \sqrt{\det I_n}} e^{\Delta_n^T I_n^{-1} \Delta_n / 2} [1 + o_{\mathfrak{E}}(1)]. \quad (28)$$

The pseudo-true parameter has displacement $O(n^{-1})$, including when $z_0 \notin \text{supp } \nu_n$. No limit of I_n is assumed.

Proof. The energy lemma proves uniform response smoothness and transient summability of every fixed order. Proposition 3.4 proves global contrast with exponential failure probability. Apply Theorem 2.9 and Proposition 2.5. $\square$

4. STRONG FILTER JETS AND THE INFERRED MEMORY KERNEL

We retain the finite-dimensional filter and memory consequences, with the ambient spaces and measurability specified independently of the data. For the homogeneous lattice, let

$$w_{n,\vartheta}^{\nu_n}(dz) = e^{V_n^\vartheta(z) - R_n^{\nu_n}(\vartheta)} \nu_n(dz), \quad \Pi_{n,\vartheta}^{\nu_n} = (z \mapsto x_n^\vartheta + e^{t_n A_\vartheta} z)_{\#} w_{n,\vartheta}^{\nu_n}.$$

Here V_n, R_n are the exact nuisance potential and its log integral from (5). All parameter differentiation holds the observed action record fixed.

For $j \geq 0$, let $\mathbb{E}_j = (C_b^{j+1}(\mathbf{H}))'$, where the test norm is the maximum of the sup norms of all Frechet derivatives through order $j+1$. Let $\mathbb{E}_j^0$ be the closed span in this dual of

$$\phi \mapsto D^\ell \phi(x)[v_1, \dots, v_\ell], \quad x, v_1, \dots, v_\ell \in \mathbf{H}, \quad 0 \leq \ell \leq j.$$

These evaluation maps are norm continuous in all their arguments: their variation in x is at most $\|x - y\| \prod_k \|v_k\|$, using the extra test derivative. Since $\mathbf{H}$ is separable, their closed linear span is a deterministic separable Banach space. Parameter tensors use operator norms on symmetric multilinear maps from $\mathbb{R}^5$.

Lemma 4.1 (Concentrated push-forward jets). *Let ν be any probability on the radius- R Hilbert ball and $p_\theta \nu$ a probability with $\theta \mapsto p_\theta$ of class C^r in $L^1(\nu)$. On a neighbourhood of a compact parameter set, suppose x_θ and S_θ are C^{r+1} , all derivatives of S through order $r+1$ have norm at most $\epsilon \leq 1$, and set*

$$K = 1 + \max_{|\alpha| \leq r+1} \sup_\theta \|\partial^\alpha x_\theta\|, \quad M_r = 1 + \max_{|\alpha| \leq r} \sup_\theta \|\partial^\alpha p_\theta\|_{L^1(\nu)}.$$

For $0 \leq j \leq r$, the push-forward has a strong order- j derivative in $\mathbb{E}_j^0$, and

$$\sup_\theta \left\| \partial_\theta^j [(x_\theta + S_\theta z)_{\#} (p_\theta \nu)] - \partial_\theta^j \delta_{x_\theta} \right\|_{\mathbb{E}_j} \leq C_r (1+R)^{r+1} K^{r+1} M_r \epsilon. \quad (29)$$

Under jointly measurable random dependence of the displayed derivatives, the dual-valued fields and their parameter suprema are measurable.

Proof. For a unit test function use

$$\phi(x_\theta + S_\theta z) - \phi(x_\theta) = \int_0^1 D\phi(x_\theta + tS_\theta z)[S_\theta z]dt.$$

Every differentiated term through order j retains one factor Sz or a derivative of it; all other factors involve bounded derivatives of x , vectors of norm at most R , and test derivatives through order $j + 1$. Leibniz's rule after multiplying by p_θ gives (29), linearly in the L^1 bound M_r . For strong differentiability, the chain rule gives finite sums of Bochner integrals of the evaluation maps defining $\mathbb{E}_j^0$. Uniform test-function Taylor bounds, the L^1 Taylor formula for p , and dominated convergence justify the derivatives in dual norm. Each integrand is strongly measurable in the same deterministic separable space. Joint measurability and integration give strong measurability in the random case. Norm continuity in θ makes the supremum equal to the supremum over a fixed countable dense parameter set. $\square$

Theorem 4.2 (Time-uniform strong filter jets). *For every fixed r and $q < \infty$, the homogeneous balanced experiment satisfies, for $0 \leq j \leq r$,*

$$\sup_{E \in \mathfrak{E}_n} E_E \left[\sup_{\vartheta \in \Theta} \|\partial_\vartheta^j \Pi_{n,\vartheta}^{\nu_n} - \partial_\vartheta^j \delta_{x_\vartheta}^q\|_{\mathbb{E}_j}^q \right] \leq C_{r,q} e^{-\gamma_{r,q} n}. \quad (30)$$

The claim holds for arbitrary Borel working priors on the fixed Hilbert ball. For two such priors, $W_1(\Pi_{n,\vartheta}^{\nu_n^1}, \Pi_{n,\vartheta}^{\nu_n^2}) \leq 2MRe^{-\lambda t_n}$ deterministically.

Proof. Differentiation of a normalized exponential tilt gives centered polynomials in derivatives of V_n . For example $\partial w = (\partial V - w(\partial V))w$. Induction, and the nuisance budget, therefore bound all fixed-order $L^1(\nu_n)$ derivatives of its density by $C_r(1 + W_{n,r,E})^{C_r}$, uniformly in ϑ ; this random bound has every finite moment. Differentiability in total variation follows from the same bounded Taylor formulas for the tilt. No density of ν_n is needed. For derivative orders above those used in the abstract BvM, repeat the nuisance argument: the affine physical generator has bounded derivatives of every fixed order.

Stability and bounded forcing bound every derivative of x_n^ϑ . Derivatives of $S_\vartheta = e^{t_n A_\vartheta}$ are at most $C_r e^{-\lambda_r t_n}$, and $t_n \geq n\underline{t}$ by the common duration lower bound. Apply the push-forward lemma and then take moments. Finitely many early n for which its $\epsilon \leq 1$ condition is not met are absorbed into $C_{r,q}$. Coupling two posterior initial states in the same radius- R ball proves the Wasserstein assertion. $\square$

Corollary 4.3 (Current-state Gaussian image). *Let $G_n = D_\vartheta x_n^{\vartheta_0} : \mathbb{R}^5 \rightarrow \mathbb{H}$. The joint parameter-initial-state posterior law of $\sqrt{n}\{z(t_n) - x_n^{\vartheta_0}\}$ is at bounded-Lipschitz distance $o_{\mathfrak{E}}(1)$ from the image under G_n of $N(I_n^{-1}\Delta_n, I_n^{-1})$. Its comparison covariance is $G_n I_n^{-1} G_n^*$, of rank at most five.*

Proof. On every fixed local parameter ball, Taylor's formula gives an error $O(|h|^2/\sqrt{n})$ from replacing $\sqrt{n}(x_n^{\vartheta_0+h/\sqrt{n}} - x_n^{\vartheta_0})$ by $G_n h$. The initial-state displacement is at most $\sqrt{n}MRe^{-\lambda t_n}$. Use the parameter total-variation theorem, root- n tightness, and these deterministic errors. The push-forward of a probability cannot increase total variation; bounded-Lipschitz test functions control the errors. $\square$

Resolve $r = (q_0, v_0)$ and let the remaining coordinates be z_Q . Writing L_D for the half-line Dirichlet Laplacian, the unresolved and resolved generators are

$$A_Q = \begin{pmatrix} 0 & I \\ -(k_b I + \epsilon L_D) & -c_b I \end{pmatrix}, \quad A_P = \begin{pmatrix} 0 & 1 \\ -(k + \epsilon) & -c \end{pmatrix}.$$

The coupling $B_Q r$ inserts ϵq_0 into the acceleration at site one, and $C_Q z_Q = (0, \epsilon q_1)$. Variation of constants gives

$$\dot{r}(t) = A_{Pr}(t) + \begin{pmatrix} 0 \\ u(t) \end{pmatrix} + \int_0^t \mathcal{K}_\vartheta(t-s)r(s)ds + C_Q e^{tA_Q} z_{Q,0}, \quad \mathcal{K}_\vartheta(t) = C_Q e^{tA_Q} B_Q. \quad (31)$$

For $0 < \omega$ smaller than a common unresolved decay rate, let $\mathcal{X}_\omega$ be the Banach space of continuous matrix kernels with norm $\sup_{t \geq 0} e^{\omega t} \|K(t)\|$. The map $\mathcal{M} : \vartheta \mapsto \mathcal{K}_\vartheta$ is smooth into this space, with bounded derivatives of every fixed order, by Duhamel differentiation and the common exponential bound. Its image and derivative images have deterministic separable closed linear spans, so the random push-forwards below are Borel laws.

For large positive real z , set $p(z) = z^2 + c_b z + k_b + 2\epsilon$. The tail Schur complement gives

$$\hat{\mathcal{K}}_\vartheta(z) = \begin{pmatrix} 0 & 0 \\ s(z) & 0 \end{pmatrix}, \quad s(z) = \{p(z) - \sqrt{p(z)^2 - 4\epsilon^2}\}/2, \quad \hat{h}_\vartheta(z) = \frac{1}{z\{z^2 + cz + k + \epsilon - s(z)\}},$$

with the square-root branch selected by $s(z) \sim \epsilon^2/z^2$.

Corollary 4.4 (Functional posterior for the memory kernel). *In $\mathcal{X}_\omega$, the posterior law of $\sqrt{n}(\mathcal{K}_\vartheta - \mathcal{K}_{\vartheta_0})$ is at bounded-Lipschitz distance $o_\epsilon(1)$ from $(D\mathcal{M}_{\vartheta_0})_\# N(I_n^{-1}\Delta_n, I_n^{-1})$. The same holds for transforms on each fixed right half-plane where the resolvents and their required derivatives are uniformly bounded.*

Proof. The bounded second derivative of $\mathcal{M}$ gives the uniform Banach-valued local remainder $O(|h|^2/\sqrt{n})$. Apply the finite-dimensional BvM and its root- n tightness. For transforms use the resolvent identity and the same Taylor argument. $\square$

5. FINITE PREPARATION MIXTURES

Let a latent label $L \in \{1, \dots, m\}$ have prior weights $w_j > 0$, and let the working initial-state law given $L = j$ be ν_j on the fixed Hilbert ball. The parameter prior is common to all labels, and the policy is not told the latent label. For one fixed infinite balanced policy, summability gives a C^r almost-sure limit $R_\infty^{\nu_j}(\vartheta) = \lim_n R_n^{\nu_j}(\vartheta)$.

Theorem 5.1 (Finite-preparation evidence mixture). *The joint label and rescaled parameter posterior is at total-variation distance $o_P(1)$ from*

$$\left(\sum_{j=1}^m \rho_{j,\infty} \delta_j \right) \otimes N(I_n^{-1}\Delta_n, I_n^{-1}), \quad \rho_{j,\infty} = \frac{w_j e^{R_\infty^{\nu_j}(\vartheta_0)}}{\sum_l w_l e^{R_\infty^{\nu_l}(\vartheta_0)}}.$$

For horizon-dependent policies use the finite-horizon weights with R_n in place of R_∞ ; no common limiting weight across different policies is required.

Proof. Apply the relative evidence expansion to each of the finitely many working priors. Since $L_n^{\nu_j}(\vartheta_0) = L_n^0(\vartheta_0) e^{R_n^{\nu_j}(\vartheta_0)}$, all other Laplace factors are common up to a relative $o_P(1)$ error. Normalization gives the weights. The nuisance potential series and its fixed-order derivatives converge uniformly by their summable envelope; positivity of the integral gives the asserted limit of R_n . Each conditional parameter posterior has the same Gaussian approximation. Total variation of the finite mixture is bounded by the sum of component errors and the ℓ^1 error of the weights. These weights are finite-component Bayes factors, not thermodynamic phases. $\square$

6. FINITE-SAMPLE LIKELIHOOD GEOMETRY FOR ADAPTIVE EXPERIMENTS

This section supplies estimates at shrinking net resolution, not just a fixed-resolution strong law. It also distinguishes a uniform probability bound from a single event asserted for uncountably many policies.

Let $\mathcal{K}$ be a compact metric parameter space with metric d . At stage i , let $\mathcal{F}_{i-1}$ be the observed past and $\mathcal{A}_i$ the information after all current actions but before the readout. The noise ξ_i , conditionally on $\mathcal{A}_i$, is $N(0, \sigma^2)$, independent of that information. Policy kernels are parameter independent on every history; candidate likelihoods hold the entire realized action record fixed. Let $f_i(\beta_0) = 0$ and assume, pathwise on all histories,

$$|f_i(\beta)| \leq M, \quad |f_i(\beta) - f_i(\gamma)| \leq d(\beta, \gamma), \quad M \geq 1. \quad (32)$$

Each field is $\mathcal{A}_i$ -measurable. Denote the logarithmic covering number by $\mathcal{H}_d(\epsilon) = \log N(\mathcal{K}, d, \epsilon)$. Define

$$Z_n = n^{-1} \sum_i |\xi_i|, \quad z = 3\sigma + 1, \quad V = \max\{1, M^4, \sigma^2 M^2\}, \quad e(t) = \min\{1, t/(8M), t/(2z)\}.$$

Proposition 6.1 (Policy-uniform shrinking-net concentration). *For every $n \geq 1$, $t > 0$, and every admissible policy, including horizon-dependent policies, the probability that either*

$$\sup_{\beta} \left| n^{-1} \sum_i \xi_i f_i(\beta) \right| > t \quad \text{or} \quad \sup_{\beta} \left| n^{-1} \sum_i \{f_i(\beta)^2 - E[f_i(\beta)^2 \mid \mathcal{F}_{i-1}]\} \right| > t$$

is at most

$$\Psi_n(t) = 4 \exp\{\mathcal{H}_d(e(t)) - nt^2/(8V)\} + e^{-n}. \quad (33)$$

All suprema are measurable. The same estimate holds uniformly over the true parameter whenever the constants and covering numbers are common.

Proof. At one deterministic net point, the centered square is a martingale difference bounded in absolute value by M^2 . Conditional exponential bounds and Markov's inequality give

$$P\{|n^{-1} \sum_i (f_i^2 - E_{i-1} f_i^2)| > t/2\} \leq 2e^{-nt^2/(8M^4)}.$$

For the score, conditioning first on $\mathcal{A}_i$ gives $E[e^{\lambda \xi_i f_i} \mid \mathcal{A}_i] = e^{\lambda^2 \sigma^2 f_i^2 / 2}$. Iterated conditioning therefore gives the corresponding bound $2e^{-nt^2/(8\sigma^2 M^2)}$. These calculations do not condition on future actions or on the final design matrix.

The elementary bound $Ee^{|\xi_i|/\sigma} \leq 2e^{1/2}$, followed by iterated conditioning, proves $P(Z_n > 3\sigma) \leq e^{n(\log 2 + 1/2 - 3)} \leq e^{-n}$. For parameters in one $e(t)$ -ball, square-field interpolation costs at most $4Me(t) \leq t/2$, including the conditional expectation; score interpolation costs at most $e(t)Z_n \leq t/2$ on $Z_n \leq z$. A union bound over the deterministic net proves (33). Continuous fields on a compact metric space have suprema over a fixed countable dense set, proving measurability. All constants are independent of the policy and sample size. $\square$

6.1. An exact integrated-likelihood remainder. Suppose the observation model is

$$Y_i = m_i^{\beta_0} + b_i^{\beta_0}(z_0) + \xi_i, \quad f_i(\beta) = m_i^\beta - m_i^{\beta_0},$$

and $\sup_{\beta, \|z\| \leq R} |b_i^\beta(z)| \leq r_i$, for a common deterministic summable sequence. Put $R_1 = \sum_i r_i$, $R_2 = \sum_i r_i^2$. The working prior ν_n may be any probability on this Hilbert ball. No density or weak convergence of ν_n is assumed.

Lemma 6.2 (Source-independent transient budget). *Writing $\ell_n(\beta) = n^{-1} \log[L_n^{\nu_n}(\beta)/L_n^{\nu_n}(\beta_0)]$, one has*

$$\ell_n(\beta) = \frac{1}{n\sigma^2} \sum_i \xi_i f_i(\beta) - \frac{1}{2n\sigma^2} \sum_i f_i(\beta)^2 + \mathcal{R}_n(\beta)/n, \quad \sup_{\beta} |\mathcal{R}_n(\beta)| \leq B_n, \quad (34)$$

where

$$B_n = \sigma^{-2} \{2 \sum_{i \leq n} r_i |\xi_i| + 3MR_1 + 3R_2\}, \quad b_0 = 3(MR_1 + R_2)/\sigma^2, \quad b_1 = 2R_1/\sigma.$$

In particular, for $u > 0$,

$$P\{B_n > b_0 + b_1\sqrt{2u}\} \leq 2e^{-u}. \quad (35)$$

Proof. Let $L_n^\star$ be the product Gaussian likelihood with means m_i^β . Gaussian completion expresses $L_n^{\nu_n}/L_n^\star$ as the integral of $\exp V_n^\beta(z)$, where

$$V_n^\beta(z) = \sigma^{-2} \sum_i \{b_i^\beta(z)(Y_i - m_i^\beta) - \frac{1}{2}b_i^\beta(z)^2\}.$$

Its absolute value is bounded by $\sigma^{-2}\{\sum r_i|\xi_i| + MR_1 + 3R_2/2\}$. A normalized positive integral has logarithm between the infimum and supremum of its exponent. Apply this at β and β_0 ; the ideal likelihood expansion has the additional term $\sigma^{-2}\sum b_i^{\beta_0}(z_0)f_i(\beta)$, bounded by MR_1/σ^2 . This proves the stated budget. For $R_1 > 0$, convexity, with weights r_i/R_1 and any remaining weight at zero, gives

$$E \exp\{\lambda \sum_{i \leq n} r_i |\xi_i|\} \leq 2 \exp\{\lambda^2 \sigma^2 R_1^2/2\}.$$

Optimizing Markov's inequality yields (35). For $R_1 = 0$ the remainder is identically zero. $\square$

Theorem 6.3 (A finite-n posterior separation inequality). *Let $F \subset \mathcal{K}$ be a measurable set satisfying, pathwise,*

$$\inf_{\beta \in F} n^{-1} \sum_i E[f_i(\beta)^2 \mid \mathcal{F}_{i-1}] \geq q > 0.$$

If the prior mass of the d -ball of radius $\sqrt{q}/4$ around the truth is at least $e^{-\mathcal{P}_d(\sqrt{q}/4)}$, then

$$\Pi_n^{\nu_n}(F) \leq \exp\{\mathcal{P}_d(\sqrt{q}/4) - nq/(4\sigma^2)\} \quad (36)$$

except on an event of probability at most

$$\Psi_n(q/16) + P\{B_n > nq/(32\sigma^2)\}, \quad (37)$$

where the second term is explicitly bounded by (35). Thus $\mathcal{P}_d$ is an upper bound on minus log prior mass.

Proof. On the good event of Proposition 6.1 with $t = q/16$, the numerator likelihood ratio over F is at most $\exp\{-13nq/(32\sigma^2) + B_n\}$. On the indicated prior ball, $|f_i| \leq \sqrt{q}/4$ at every stage, and the same score bound gives a likelihood ratio at least $\exp\{-3nq/(32\sigma^2) - B_n\}$. Divide the integrated numerator by the integrated denominator and use $2B_n \leq nq/(16\sigma^2)$. The remaining exponent is $-nq/(4\sigma^2) + \mathcal{P}_d$, as claimed. Every coefficient in this calculation is retained; there is no unspecified constant multiplying a shrinking separation. $\square$

6.2. Triangular experiments and nonstationary Laplace limits. For a fixed infinite policy, the usual noise coupling is available. For horizon-dependent policies, choose any joint probability space carrying one admissible row of length n for each n ; rows need not be independent. For each fixed $t > 0$, the right side of (33) is summable in n . So is the transient probability at a fixed positive multiple of n . Borel–Cantelli at countably many $t \downarrow 0$ proves, for each such chosen policy sequence and coupling,

$$\|\ell_n + Q_n\|_\infty \longrightarrow 0 \quad \text{almost surely,} \quad Q_n(\beta) = \frac{1}{2n\sigma^2} \sum_i E[f_i(\beta)^2 \mid \mathcal{F}_{i-1}]. \quad (38)$$

The statement is not an intersection of probability-one events over an uncountable collection of policies. Uniformity means that the finite-n bounds preceding it are common to the full class.

Theorem 6.4 (Adaptive posterior Laplace tracking and cluster LDPs). *Assume the preceding hypotheses and a fixed full-support prior on $\mathcal{K}$. For every continuous $\varphi : \mathcal{K} \rightarrow \mathbb{R}$,*

$$-\frac{1}{n} \log \int e^{-n\varphi(\beta)} \Pi_n^{\nu_n}(d\beta) - \inf_{\beta \in \mathcal{K}} \{\varphi(\beta) + Q_n(\beta)\} \longrightarrow 0 \quad \text{almost surely.} \quad (39)$$

The sequence Q_n is precompact in the uniform norm. Every uniform cluster limit I is a good rate function, vanishes at β_0 , and the corresponding subsequence of posteriors satisfies the full open-set lower and closed-set upper LDP with rate I and speed n . If $Q_n \rightarrow I$ uniformly along the full sequence, this is a full adaptive posterior LDP. These statements hold on one event for all continuous φ and all Borel sets.

Proof. One has $0 \leq Q_n \leq M^2/(2\sigma^2)$, $Q_n(\beta_0) = 0$, and $|Q_n(\beta) - Q_n(\gamma)| \leq Md(\beta, \gamma)/\sigma^2$. Arzela–Ascoli gives precompactness. For every radius $s > 0$, full support and a finite $s/2$ -cover give

$$m(s) := \inf_{\beta \in \mathcal{K}} \Pi\{d(\gamma, \beta) < s\} > 0.$$

For the equicontinuous family $\varphi + Q_n$, integration over an s -ball about a minimizer gives

$$\inf(\varphi + Q_n) \leq -n^{-1} \log \int e^{-n(\varphi + Q_n)} d\Pi \leq \inf(\varphi + Q_n) + \omega_\varphi(s) + Ms/\sigma^2 - n^{-1} \log m(s).$$

Use this direct upper/lower integral bound also with $\varphi = 0$, use (38), then let $n \rightarrow \infty$ and $s \downarrow 0$. The argument is deterministic once the uniform tracking event holds, so it holds there for every continuous function without another exceptional set. On a subsequence with $Q_n \rightarrow I$, it gives the uniform log-likelihood limit $-I$. For an open set integrate a small neighbourhood of any one of its points; for a closed set take the supremum of the likelihood over that compact set. The denominator has exponent zero by the preceding argument. These give the two LDP bounds simultaneously for all Borel sets. $\square$

7. COMPLETE RECOVERY OF AN UNKNOWN INFINITE JACOBI BATH

Fix $0 < c_- < c_+ < \infty$, $0 < a_- < a_+ < \infty$, and $2a_+ < b_- < b_+ < \infty$. The compact metrizable parameter space is

$$\mathfrak{B} = [c_-, c_+] \times \prod_{j \geq 0} [a_-, a_+] \times \prod_{j \geq 0} [b_-, b_+], \quad \beta = (c, (a_j), (b_j)). \quad (40)$$

On $\ell^2(\mathbb{N}_0)$, let

$$(J_\beta q)_0 = b_0 q_0 - a_0 q_1, \quad (J_\beta q)_j = b_j q_j - a_{j-1} q_{j-1} - a_j q_{j+1}, \quad j \geq 1.$$

The operator satisfies $(b_- - 2a_+)I \leq J_\beta \leq (b_+ + 2a_+)I$. Put

$$A_\beta = \begin{pmatrix} 0 & I \\ -J_\beta & -cI \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ e_0 \end{pmatrix}, \quad \ell(q, v) = q_0, \quad h_\beta(t) = \ell \int_0^t e^{sA_\beta} B ds.$$

Lemma 7.1 (Product continuity and common stability). *The map $\beta \mapsto h_\beta|_{[0, T]}$ is continuous into $C([0, T])$. There are common constants $C_* \geq 1$, $\lambda_* > 0$ with $\sup_\beta \|e^{tA_\beta}\| \leq C_* e^{-\lambda_* t}$.*

Proof. Apply the energy proof of Lemma 3.1 with $C = cI$ and $K = J_\beta$. Coordinatewise convergence gives convergence of $J_\beta q$ on finitely supported vectors. Uniform operator bounds and density extend this to strong convergence on ℓ^2 . Powers and uniformly convergent exponential series then give strong semigroup convergence uniformly on compact time intervals. Apply to B , observe with ℓ , and integrate. $\square$

7.1. Weyl reconstruction. For the tail starting at j , define $m_j(w) = \langle e_j, (w + J_\beta^{(j)})^{-1} e_j \rangle$, $w > 0$. Laplace transformation and a Schur complement give

$$\hat{h}_\beta(z) = z^{-1} m_0(z^2 + cz), \quad c = -h_\beta'''(0), \quad m_j(w)^{-1} = w + b_j - a_j^2 m_{j+1}(w). \quad (41)$$

Consequently

$$b_j = \lim_{w \rightarrow \infty} \{m_j(w)^{-1} - w\}, \quad a_j^2 = \lim_{w \rightarrow \infty} w \{w + b_j - m_j(w)^{-1}\}, \quad m_{j+1}(w) = \{w + b_j - m_j(w)^{-1}\} / a_j^2. \quad (42)$$

Theorem 7.2 (Complete Jacobi reconstruction). *The boundary response determines all of β . The infinite-chain Weyl function is nonrational, so no fixed finite Jacobi chain has the same response.*

Proof. Recover damping from the third derivative. The map $z \mapsto z^2 + cz$ is a bijection of $(0, \infty)$, so the Laplace transform determines $m_0(w)$ there. Since $m_{j+1}(w) = w^{-1} + O(w^{-2})$, (42) recovers b_j, a_j, m_{j+1} successively; take the positive square root for a_j . The vector e_0 is cyclic: the recurrence and $a_j > 0$ inductively put every basis vector in the span of $\{J_\beta^k e_0\}$. A rational Stieltjes transform has a finite-support spectral measure and therefore a finite-dimensional cyclic subspace, a contradiction. The inverse spectral recursion is classical [1]; the finite condition estimates used statistically are proved separately below. $\square$

For a fixed dense duration law ω on $(0, T]$, define

$$D(\beta, \gamma) = \sum_m \omega_m [h_\beta(t_m) - h_\gamma(t_m)]^2, \quad d_J(\beta, \gamma) = \max\{|c - \tilde{c}|, \max_{j \leq J} |a_j - \tilde{a}_j|, \max_{j \leq J} |b_j - \tilde{b}_j|\}. \quad (43)$$

The square root of D induces the product topology: the response map is a continuous injection from a compact space into a Hausdorff space. Injectivity on $[0, T]$ follows from bounded-generator analyticity and the reconstruction theorem. Hence, whenever the constraint is nonempty,

$$\kappa_J(\delta) = \inf_{d_J(\beta, \gamma) \geq \delta} D(\beta, \gamma) > 0. \quad (44)$$

The effective lower bound in Theorem 9.5 will replace this qualitative compactness argument in every rate assertion.

8. THE EXACT POSTERIOR UNDER LOGARITHMIC WASHOUT

At stage i , a parameter-independent bounded feedback segment of length at most T_J and force at most U_J is followed by zero force for

$$w_i = \frac{1 + \epsilon_w}{\lambda_*} \log(i + 1), \quad \epsilon_w > 0. \quad (45)$$

Then draw $M_i \sim \omega$, draw an independent Rademacher S_i , apply aS_i for duration t_{M_i} , and finally draw the independent $N(0, \sigma^2)$ readout. All constants are common across policies and horizons. The initial state lies in the fixed radius- R Hilbert ball. Writing $U_* = \max(U_J, a)$, common stability gives

$$Y_i = aS_i h_{\beta_0}(t_{M_i}) + \eta_i^{\beta_0}(z_0) + \xi_i, \quad \sup_{\beta, \|z\| \leq R} |\eta_i^\beta(z)| \leq r_i := C_*^2(R + U_*/\lambda_*)(i + 1)^{-1 - \epsilon_w}. \quad (46)$$

Indeed the state preceding washout has norm at most $C_*(R + U_*/\lambda_*)$, and propagation through washout and the probe multiplies this by at most $C_* e^{-\lambda_* w_i}$.

Lemma 8.1 (All-order washout differentiation at the original decay rate). *For each fixed finite-cylinder multi-index α ,*

$$\sup_{\beta, \|z\| \leq R} |\partial^\alpha \eta_i^\beta(z)| \leq C_\alpha (1 + \log(i + 1))^{|\alpha|} (i + 1)^{-1 - \epsilon_w}. \quad (47)$$

For every $0 < \bar{\epsilon} < \epsilon_w$, this is bounded by $C_{\alpha, \bar{\epsilon}}(i + 1)^{-1 - \bar{\epsilon}}$, hence is summable.

Proof. The generator is affine in cylinder coordinates. An order- r Duhamel derivative of e^{wA_β} is a sum of at most $r!$ ordered integrals. Each integrand is bounded by $C_*^{r+1} D^r e^{-\lambda_* w}$, because the durations of its $r + 1$ semigroup factors add to w ; D bounds the relevant generator derivatives. The simplex volume is $w^r/r!$. Thus the derivative is bounded by $C_r w^r e^{-\lambda_* w}$, retaining the same λ_* . Derivatives of the pre-washout state are uniformly bounded by integration of $C_r(1 + t)^r e^{-\lambda_* t}$ against the bounded realized input; all actions are held fixed. Derivatives during the bounded probe interval are bounded as well. Leibniz's rule therefore gives $C_\alpha(1 + w_i)^{|\alpha|} e^{-\lambda_* w_i}$. Substitute (45). Finally every fixed power of $1 + \log x$ is bounded by a constant times $x^{\epsilon_w - \bar{\epsilon}}$ for $x \geq 1$. $\square$

Let $L_n^{\nu_n}$ be the exact integrated likelihood of the full realized action record, with any normalized working initial-state prior ν_n on the same Hilbert ball. Let Π be a fixed Borel prior with full support on $\mathfrak{B}$. It defines the exact working posterior $\Pi_n^{\nu_n}$ by normalization of $L_n^{\nu_n} d\Pi$.

Theorem 8.2 (Full good posterior LDP). *For every admissible policy sequence and every chosen triangular coupling as specified in Section 6, on one event of probability one, for every Borel set $A \subset \mathfrak{B}$,*

$$-\inf_{A^\circ} I_{\beta_0} \leq \liminf_n n^{-1} \log \Pi_n^{\nu_n}(A) \leq \limsup_n n^{-1} \log \Pi_n^{\nu_n}(A) \leq -\inf_A I_{\beta_0}, \quad I_{\beta_0}(\beta) = \frac{a^2}{2\sigma^2} D(\beta, \beta_0). \quad (48)$$

This includes any fixed infinite admissible policy. Finite- n probability bounds are uniform over all policies with the declared common bounds.

Proof. Use $f_i(\beta) = aS_i[h_\beta(t_{M_i}) - h_{\beta_0}(t_{M_i})]$ and $d(\beta, \gamma) = a\|h_\beta - h_\gamma\|_\infty$ in Proposition 6.1. This is a compact metric geometry, and $|f_i| \leq 2aC_*/\lambda_*$. Conditional squared means are exactly $a^2 D(\beta, \beta_0)$. The summable residual in (46) meets Lemma 6.2, including the candidate residual from every past forcing segment. Hence

$$\sup_\beta |n^{-1} \log[L_n^{\nu_n}(\beta)/L_n^{\nu_n}(\beta_0)] + I_{\beta_0}(\beta)| \rightarrow 0$$

almost surely by the summable finite- n bounds. Only undifferentiated residuals are needed here. The posterior normalizer has exponent zero: its upper bound follows from $I \geq 0$, and integration over the open set $\{I < \eta\}$, of positive prior mass, gives lower exponent at least $-\eta$. For an open set integrate a small neighbourhood of any of its points; for a closed set use the uniform likelihood upper bound and compactness. Subtract the normalizer exponent. This proves the bounds simultaneously for all sets, as in Theorem 6.4. $\square$

Corollary 8.3 (Strong consistency and effective cylinder exponent). *For every product neighbourhood U of β_0 , $\Pi_n^{\nu_n}(U^c) \rightarrow 0$ almost surely. Under the multiscale design of (62),*

$$\limsup_n n^{-1} \log \Pi_n^{\nu_n}\{d_J(\beta, \beta_0) \geq \delta\} \leq -\frac{a^2}{2\sigma^2} \kappa_J(\delta).$$

Proof. The continuous good rate is positive off its unique zero. Apply the closed upper bound and, for cylinders, the effective inverse-stability theorem. $\square$

The physical clock satisfies

$$\mathsf{T}_n = \kappa_w n \log n + O(n), \quad \kappa_w = (1 + \epsilon_w)/\lambda_*, \quad v(t) = \frac{t}{\kappa_w W(t/\kappa_w)}, \quad v(\mathsf{T}_n)/n \rightarrow 1, \quad (49)$$

where W is the principal Lambert function. This follows by summing w_i and using $\log((n+1)!) = n \log n + O(n)$; other stage durations are uniformly bounded. Thus the same LDP holds at elapsed-time speed $v(\mathsf{T}_n)$.

9. EFFECTIVE JACOBI INVERSION WITH A COMPLETE CONDITION CALCULATION

Throughout this section, vectors have the maximum norm unless an explicit ℓ^1 norm is written; matrix infinity norms are induced row-sum norms. Set

$$B_0 = \max\{1, b_+ + 2a_+\}, \quad \Lambda = 1 + c_+ + B_0, \quad Q = 8 \max\{2, \Lambda, T, T^{-1}, a_-^{-1}\}.$$

For a depth $J \geq 0$, write $j = J + 1$ and define

$$R_J = 4J + 8, \quad L_J = Q^{64(J+1)^2}. \quad (50)$$

All constants below are arithmetic expressions in the declared coefficient box and probe horizon. In particular, none is an infimum of an inverse map.

9.1. Response derivatives, moments, and finite Gram systems. Put $\mu_k = \langle e_0, J_\beta^k e_0 \rangle$ and $v_r = h_\beta^{(r+2)}(0)$. Then $\mu_0 = v_0 = 1$ and $c = -v_1$. The large-frequency resolvent expansion gives the convergent identity

$$\sum_{r \geq 0} v_r x^r = \langle e_0, (I + cx + x^2 J_\beta)^{-1} e_0 \rangle = \sum_{k \geq 0} (-1)^k \mu_k x^{2k} (1 + cx)^{-k-1}. \quad (51)$$

Indeed, factor out $1 + cx$, expand the remaining bounded-operator geometric series near zero, and apply the boundary vector. Comparing the coefficient of x^{2m} yields

$$\mu_m = (-1)^m \left[v_{2m} - \sum_{k=0}^{m-1} (-1)^k \binom{2m-k}{2m-2k} c^{2m-2k} \mu_k \right]. \quad (52)$$

This also proves exact response-moment triangularity; no analytic continuation is used in the finite calculation.

Lemma 9.1 (Moment-map condition bound). *For $M \geq 1$, two admissible parameter values satisfy*

$$\max \left\{ |c - \tilde{c}|, \max_{k \leq M} |\mu_k - \tilde{\mu}_k| \right\} \leq Q^{4(M+1)^2} \max_{0 \leq r \leq 2M} |v_r - \tilde{v}_r|. \quad (53)$$

For $M = 0$, include v_1 in the input maximum to recover damping.

Proof. Let $S = Q/8 \geq 2$, and denote the input difference by e . Set $P_0 = 0$ and, for $m \geq 1$,

$$P_m = 1 + \sum_{k < m} \binom{2m-k}{2m-2k} \{ S^{2m-2k} P_k + (2m-2k) S^{2m-k-1} \}.$$

The identity $|x^r - y^r| \leq r S^{r-1} |x - y|$, the bound $|\mu_k| \leq B_0^k \leq S^k$, and (52) prove $|\mu_m - \tilde{\mu}_m| \leq P_m e$ by induction. With $U_m = \max(1, P_0, \dots, P_m)$,

$$U_m \leq [1 + m(1 + 2m) 4^m S^{2m}] U_{m-1} \leq Q^{4m} U_{m-1}.$$

For the last inequality, $m(1 + 2m) \leq 16^m$ (induction, starting at $m = 1$) and $1 + 64^m S^{2m} \leq 2Q^{2m} \leq Q^{4m}$. Thus $U_M \leq Q^{2M(M+1)} \leq Q^{4(M+1)^2}$. Damping costs only the factor one because $c = -v_1$. This specifies every operation and norm. $\square$

Let $H_j = (\mu_{r+s})_{0 \leq r, s \leq j}$, $\Delta_j = \det H_j$, and $\Delta_{-1} = 1$. For $j \geq 1$ solve

$$H_{j-1} \alpha^{(j)} = -(\mu_j, \dots, \mu_{2j-1})^T, \quad p_j(x) = x^j + \sum_{r < j} \alpha_r^{(j)} x^r, \quad p_0 = 1. \quad (54)$$

Write $\rho_j = p_j^T H_j p_j$ and $t_j = \sum_{r, s \leq j} p_{j,r} p_{j,s} \mu_{r+s+1}$. Orthogonality gives

$$b_j = t_j / \rho_j, \quad a_j^2 = \rho_{j+1} / \rho_j, \quad \rho_j = \prod_{r < j} a_r^2, \quad \Delta_j = \prod_{r < j} a_r^{2(j-r)}. \quad (55)$$

Here p_j is the monic orthogonal polynomial of the boundary spectral measure. The three-term recurrence proves $p_j(x) = \det(xI - J_{\beta, [0, j-1]})$; consequently its roots lie in $[0, B_0]$, and its coefficient vector obeys

$$\|p_j\|_1 \leq (1 + B_0)^j =: S_j. \quad (56)$$

The recurrence also proves the norm and determinant products in (55): starting with $\rho_0 = 1$, orthogonality implies $\rho_{j+1} = a_j^2 \rho_j$, and the unitriangular change from monomials to monic polynomials gives $\det H_j = \prod_{r \leq j} \rho_r$.

Lemma 9.2 (Finite-map condition ledger). *Define, for $j \geq 1$,*

$$\begin{aligned} A_j &= j! B_0^{2j(j-1)} a_-^{j(j-1)}, & V_j &= j A_j (1 + j S_j), \\ R_j &= S_j^2 + 2 B_0^{2j} S_j V_j, & T_j &= S_j^2 + 2 B_0^{2j+1} S_j V_j, & l_j &= a_-^{2j}. \end{aligned}$$

Set $S_0 = l_0 = T_0 = 1$, $V_0 = R_0 = 0$. For two admissible moment vectors at maximum distance e , the following are valid:

$$\begin{aligned} \|H_{j-1}^{-1}\|_\infty &\leq A_j, & \|p_j - \tilde{p}_j\|_1 &\leq V_j e, \\ |\rho_j - \tilde{\rho}_j| &\leq R_j e, & |t_j - \tilde{t}_j| &\leq T_j e, \\ |b_j - \tilde{b}_j| &\leq (T_j + b_+ R_j) e / l_j, & |a_j - \tilde{a}_j| &\leq (R_{j+1} + a_+^2 R_j) e / (2a_- l_j). \end{aligned} \quad (57)$$

Only moments through $2J + 2$ are required through depth J , and

$$\max_{r \leq J} (|a_r - \tilde{a}_r| \vee |b_r - \tilde{b}_r|) \leq Q^{17(J+1)^2} \max_{k \leq 2J+2} |\mu_k - \tilde{\mu}_k|. \quad (58)$$

Proof. Each cofactor of the j -by- j matrix H_{j-1} is bounded by $(j-1)! B_0^{2j(j-1)}$, using the determinant expansion and the entry bound B_0^{2j} . Its determinant is at least $a_-^{j(j-1)}$. Summing j cofactors in each inverse row proves the first bound. Subtract the two linear systems, using the first inverse and the second solution:

$$\|\alpha^{(j)} - \tilde{\alpha}^{(j)}\|_\infty \leq A_j (e + j e \|\tilde{\alpha}^{(j)}\|_\infty) \leq A_j (1 + j S_j) e.$$

Multiplication by j gives the stated coefficient ℓ^1 bound. For quadratic forms, the moment-matrix difference contributes at most $S_j^2 e$; changing the two polynomial factors contributes at most $2B_0^{2j} S_j V_j e$, or $2B_0^{2j+1} S_j V_j e$ for the shifted matrix. These are exactly R_j, T_j . Since $\rho_j \geq l_j$ and $t_j = b_j \rho_j$, subtracting the ratios gives the bound for b_j . Subtract ρ_{j+1}/ρ_j , use $a_j^2 \leq a_+^2$, and divide by $a_j + \tilde{a}_j \geq 2a_-$ to obtain the last bound.

For clarity, the all-depth exponent propagation is displayed separately. For every integer $j \geq 1$, $j! \leq 2^{j^2}$, $j \leq Q^j$, $B_0, a_+, a_-^{-1} \leq Q$, and $1 + B_0 \leq Q$. Therefore

quantity	S_j	A_j	V_j	R_j	T_j
upper bound	Q^j	Q^{4j^2}	Q^{8j^2}	Q^{13j^2}	Q^{14j^2}

Indeed the exponents before simplification for V_j, R_j, T_j are, respectively, $4j^2 + 3j + 1$, $8j^2 + 3j + 2$, and $8j^2 + 3j + 3$; each is bounded by the corresponding table entry for $j \geq 1$. The two ratio constants in (57) are at most Q^{17j^2} and $Q^{15(j+1)^2}$. At $j = 0$, the first is one and the second is at most Q^{14} , by the stated conventions. Taking the maximum through J gives (58). No constants are multiplied over unnecessary reconstruction levels. $\square$

Theorem 9.3 (All-depth effective response-jet inverse). *For all $J \geq 0$ and $\beta, \gamma \in \mathfrak{B}$,*

$$d_J(\beta, \gamma) \leq L_J \max_{0 \leq r \leq R_J} |h_\beta^{(r)}(0) - h_\gamma^{(r)}(0)|. \quad (59)$$

Proof. Take $M = 2J + 2$ in the moment lemma and compose with the Gram lemma. Derivatives through $4J + 6$ suffice, hence those through R_J suffice. For $j = J + 1 \geq 1$, the total exponent is bounded by

$$4(2j + 1)^2 + 17j^2 \leq 36j^2 + 17j^2 = 53j^2 \leq 64j^2.$$

This calculation includes every shallow depth, without an exceptional-case adjustment or a change of the base Q . $\square$

9.2. Finite Taylor grids and the sampling law. For $R \geq 1$, set

$$V_R = (k^r / r!)_{0 \leq k, r \leq R}, \quad C_R = \|V_R^{-1}\|_\infty, \quad M_R = 2e^{\Lambda^T} \Lambda^R, \quad H_R = \frac{C_R M_R R^{R+1}}{(R+1)!}.$$

The rational number C_R satisfies

$$C_R \leq 2^{R+1} (R+1)!. \quad (60)$$

To see this, the coefficient sum of the Lagrange numerator $\prod_{l \neq k} (x - l)$ is at most $(R + 1)!/(k + 1)$. The denominator has absolute value $k!(R - k)!$; multiply coefficients by $r! \leq R!$ and sum over k , using $\sum_k 1/(k!(R - k)!) = 2^R/R!$.

Lemma 9.4 (Finite-grid derivative certificate). *For $g = h_\beta - h_\gamma$ and $0 < \varepsilon \leq \min(1, T/R)$,*

$$\max_{r \leq R} |g^{(r)}(0)| \leq C_R \varepsilon^{-R} \max_{0 \leq k \leq R} |g(k\varepsilon)| + H_R \varepsilon. \quad (61)$$

The value $g(0) = 0$ is deterministic; precisely R positive-time values, not $R + 1$ measured cells, are required.

Proof. Since $h^{(r)}(t) = \ell A^{r-1} e^{tA} B$ for $r \geq 1$, $\sup_{t \leq T} |g^{(R+1)}(t)| \leq M_R$. Taylor expansion at the grid gives $V_R(\varepsilon^r g^{(r)}(0))_r$ plus a remainder of maximum norm at most $M_R(R\varepsilon)^{R+1}/(R+1)!$. Invert the matrix and use $\varepsilon^{-r} \leq \varepsilon^{-R}$. $\square$

Use labeled diagnostic cells, allowing distinct labels at the same time. Assign mass 2^{-m-1} , $m \geq 1$, to a fixed dense enumeration of rational points of $(0, T]$, and define additional cells

$$\begin{aligned} q &= (J, s, k), \quad J \geq 0, \quad s \geq 1, \quad 1 \leq k \leq R_J, \\ \varepsilon_{J,s} &= \zeta_{R_J} 2^{-s}, \quad \zeta_R = \min\{1, T/(2R)\}, \\ t_q &= k\varepsilon_{J,s}, \quad \omega_q = 2^{-J-s-2}/R_J. \end{aligned} \quad (62)$$

Each family has total mass one half. Duplicate times are summed in the response metric; there is no zero-time sampling cell. For $0 < \delta \leq 1$, let $s_J(\delta)$ be the least positive integer with $H_{R_J} \varepsilon_{J,s} \leq \delta/(2L_J)$. Define

$$\kappa_J(\delta) = \frac{2^{-J-s_J(\delta)-2}}{R_J} \left(\frac{\delta \varepsilon_{J,s_J(\delta)}^{R_J}}{2C_{R_J} L_J} \right)^2, \quad (63)$$

$$Z_J = \max\{2/\zeta_{R_J}, 4L_J H_{R_J}\}, \quad K_J = \frac{2^{-J-4}}{R_J \zeta_{R_J} C_{R_J}^2 L_J^2 Z_J^{2R_J+1}}. \quad (64)$$

Theorem 9.5 (Effective depth-dependent inverse stability). *For $d_J(\beta, \gamma) \geq \delta$,*

$$D(\beta, \gamma) \geq \kappa_J(\delta) \geq K_J \delta^{2R_J+3} \geq e^{-A_0(J+1)^3} \delta^{20(J+1)}, \quad (65)$$

where the explicit box constant is

$$A_0 = 5000\{1 + \Lambda T + |\log T| + \log Q\}. \quad (66)$$

Proof. The jet inverse and Taylor certificate force one positive grid value to have magnitude at least $\delta \varepsilon_{J,s}^{R_J}/(2C_{R_J} L_J)$. Its weighted square is the first bound. Minimality of s gives $\varepsilon_{J,s} \geq \delta/Z_J$: this is immediate when $s = 1$, and otherwise $2H_{R_J} \varepsilon_{J,s} > \delta/(2L_J)$. Substitution of $2^{-s} = \varepsilon_{J,s}/\zeta_{R_J}$ proves the K_J bound.

Here is the full logarithmic audit. Put $j = J + 1$ and $F = 1 + \Lambda T + |\log T| + \log Q$. Since $R_J \leq 8j$, $R_J + 1 \leq 9j$, and $\log x \leq x$,

$$\log C_{R_J} \leq 90j^2 F, \quad \log \zeta_{R_J}^{-1} \leq 17jF, \quad \log H_{R_J} \leq 171j^2 F, \quad \log Z_J \leq 240j^2 F.$$

The third inequality follows by dropping the negative factorial term and adding $90j^2 F + F + 8jF + 72j^2 F$; the fourth follows by adding $\log 4 + 64j^2 \log Q$ to the third and comparing with $\log 2 + 17jF$. Finally $\log \zeta_{R_J} \leq 0$, so

$$-\log K_J \leq 12jF + 308j^2 F + (16j + 1)240j^2 F \leq 4400j^3 F < A_0 j^3.$$

Also $2R_J + 3 = 8J + 19 \leq 20j$. This proves the last bound for $0 < \delta \leq 1$, with every depth and every constant accounted for. $\square$

10. NO-WASHOUT IDENTIFICATION UNDER ARBITRARY BOUNDED ADAPTIVE BASELINES

Put $g_\beta(t) = \ell e^{tA_\beta} B = h'_\beta(t)$. Product convergence gives uniform convergence of g_β on every bounded time interval, and common stability gives the tail envelope $C_* e^{-\lambda_* t}$. Dominated convergence therefore proves continuity of $\beta \mapsto g_\beta$ into $L^1(0, \infty)$, with compact image. For all realized inputs with $\|u\|_\infty \leq U_*$,

$$\sup_{t \geq 0} |((g_\beta - g_\gamma) * u)(t)| \leq U_* \|g_\beta - g_\gamma\|_1. \quad (67)$$

This holds for arbitrary switching and feedback histories, not just for a particular controller.

Fix $\tau_g > 0$, $T_0, U_0 < \infty$, and $\rho \in (0, 1]$. At stage i , the controller first chooses a baseline segment of duration $r_i \in [\tau_g, \tau_g + T_0]$. A fresh exploration flag has probability ρ ; on exploration draw a cell $M_i \sim \omega$, otherwise the controller selects any cell. The baseline during both the first segment and the final probe is bounded by U_0 , and is fixed before the fresh sign. Then add aS_i during the final interval t_{M_i} , and draw readout noise only at its end. Any baseline randomizers are included before the sign. The parameter-independent policy may depend on the horizon. Thus

$$\mathcal{F}_{i-1} \longrightarrow (r_i, u_i^0, E_i, M_i) \longrightarrow S_i \longrightarrow \xi_i \longrightarrow Y_i, \quad \tau_g n \leq \mathsf{T}_n \leq \tau_+ n, \quad \tau_+ = \tau_g + T_0 + T. \quad (68)$$

No washout or state reset is required. Write $U_* = U_0 + a$.

10.1. A deterministic metric controlling every policy. Define the compact metric

$$d_*(\beta, \gamma) = U_* \|g_\beta - g_\gamma\|_1 + a \|h_\beta - h_\gamma\|_{C([0, T])}. \quad (69)$$

Continuity follows from the compact response images. Injectivity follows from boundary reconstruction, so this metric induces the product topology. For $f_i(\beta) = m_i^\beta - m_i^{\beta_0}$, where m_i is the mean from zero initial state under the complete realized input, linearity gives

$$f_i(\beta) = P_i(\beta) + aS_i[h_\beta(t_{M_i}) - h_{\beta_0}(t_{M_i})], \quad (70)$$

with P_i measurable before S_i . Removing only the current signed pulse leaves an input bounded by U_* , hence

$$|P_i(\beta) - P_i(\gamma)| \leq U_* \|g_\beta - g_\gamma\|_1, \quad |f_i(\beta) - f_i(\gamma)| \leq d_*(\beta, \gamma).$$

The full-input convolution also gives $|f_i| \leq 2U_* C_*/\lambda_*$. Use $M = \max(1, 2U_* C_*/\lambda_*)$ in the finite-n field estimates.

Lemma 10.1 (Predictable-intercept information floor). *Conditioning on every action selected before S_i ,*

$$E[f_i(\beta)^2 \mid \mathcal{F}_{i-1}, r_i, u_i^0, E_i, M_i] = P_i(\beta)^2 + a^2 [h_\beta(t_{M_i}) - h_{\beta_0}(t_{M_i})]^2.$$

Consequently,

$$E[f_i(\beta)^2 \mid \mathcal{F}_{i-1}] \geq \rho a^2 D(\beta, \beta_0). \quad (71)$$

The same sign identity holds for directional derivatives on any fixed finite coefficient cylinder.

Proof. The cross term has mean zero under the independent fresh sign. On an exploration stage the remaining response square averages to $a^2 D$. All other conditional-square terms are nonnegative. Differentiation of the linear decomposition proves the cylinder statement. $\square$

The exact initial-state contribution at stage i is $b_i^\beta(z) = \ell e^{\mathsf{T}_i A_\beta z}$, and is bounded by $r_i = C_* R e^{-\lambda_* \tau_g i}$. The finite-n correction lemma applies with, explicitly,

$$R_1 = \frac{C_* R}{e^{\lambda_* \tau_g} - 1}, \quad R_2 = \frac{C_*^2 R^2}{e^{2\lambda_* \tau_g} - 1}.$$

Every fixed cylinder derivative is bounded by a polynomial in i times this exponential, by the same-rate Duhamel calculation.

Theorem 10.2 (Policy-uniform no-washout posterior exponent). *For each admissible policy sequence, with the coupling convention of Section 6, almost surely for every closed $F \subset \mathfrak{B}$,*

$$\limsup_n n^{-1} \log \Pi_{n,\text{lin}}^{\nu_n}(F) \leq -\frac{\rho a^2}{2\sigma^2} \inf_{\beta \in F} D(\beta, \beta_0). \quad (72)$$

For a coefficient cylinder the infimum is at least $\kappa_J(\delta)$. In physical time the guaranteed upper exponent is divided by at most τ_+ . At every finite n , the field error is bounded by (33) with the explicit metric d_ , and the exact-likelihood correction by (35), uniformly over all such policies.*

Proof. All hypotheses of Proposition 6.1 have just been checked with a deterministic compact metric common to every history and policy. The conditional-square lower bound is (71); the transient budget is the summable global initial-state envelope above. Thus uniform likelihood tracking (38) holds, with $Q_n \geq \rho a^2 D / (2\sigma^2)$. To normalize, a d_* -ball of radius s about the truth satisfies $|f_i| \leq s$ for all stages and policies. The finite- n denominator argument of Theorem 6.3, or the uniform Laplace bound with $\varphi = 0$, makes its exponent zero as $s \downarrow 0$. Integrating the likelihood upper bound over F proves the result. The effective inverse bound gives the cylinder claim. Since $\log \Pi(F) \leq 0$ and $n/\tau_n \geq 1/\tau_+$, the physical-time conclusion follows with the stated direction of inequality. $\square$

The theorem is a guaranteed information floor. The actual adaptive information Q_n also includes the baseline response. Rather than identify it incorrectly with the floor, Theorem 6.4 gives exact Laplace tracking and full LDPs along every uniformly convergent information subsequence. A matching full-sequence LDP follows whenever that predictable information converges uniformly. No convergence of the information is assumed for the universal upper exponent.

For the washed experiment one may similarly draw a fresh exploration flag, use ω on exploration, and choose all other diagnostic cells predictably before the fresh sign. With $d = a\|h_\beta - h_\gamma\|_\infty$, the same argument gives

$$\limsup_n n^{-1} \log \Pi_n^{\nu_n}(F) \leq -\frac{\rho a^2}{2\sigma^2} \inf_F D. \quad (73)$$

This is the exploration-floor extension; the finite- n and growing-depth versions are included in the next section.

11. INCREASING-DEPTH RECOVERY AND SIMULTANEOUS CONFIDENCE CYLINDERS

The following argument uses the finite-sample likelihood inequality, including its constants, at the shrinking separation supplied by the inverse theorem. The prior-thickness requirement is imposed only for these quantitative conclusions, not for the full-support posterior LDP.

11.1. Effective covers and prior mass. Assume a product prior on $c, (a_j), (b_j)$, with all coordinate densities bounded below by a common $p_- > 0$. Let $w_- > 0$ be the minimum of the three interval lengths and $W \geq 1$ their maximum, enlarged to one. Use either the washed response metric $d_h(\beta, \gamma) = a\|h_\beta - h_\gamma\|_{C([0,T])}$, or the metric d_* of (69). The following construction specifies covering numbers and prior-mass bounds for both metrics.

Lemma 11.1 (Boundary locality and explicit finite-cylinder bounds). *If two parameters agree in $c, a_0, b_0, \dots, a_K, b_K$, then*

$$\begin{aligned} \|h_\beta - h_\gamma\|_{C([0,T])} &\leq 2Te^{\Lambda T} \frac{(\Lambda T)^{K+1}}{(K+1)!}, \\ \|g_\beta - g_\gamma\|_{L^1(0,\infty)} &\leq 2Se^{\Lambda S} \frac{(\Lambda S)^{K+1}}{(K+1)!} + 2C_* e^{-\lambda_* S} / \lambda_* \quad (S > 0). \end{aligned} \quad (74)$$

If only those coordinates are changed by at most r , with the tail unchanged, the additional error in d_h is at most $C_h r$, where $C_h = \max(1, 2aT^2 e^{\Lambda T})$; the additional error in d_ is at most $C_*^{\text{met}} r$, where*

$$C_*^{\text{met}} = \max\{1, 4(U_* + a)C_*^2/\lambda_*^2\}.$$

Proof. The matrix entries of A_β only connect a site to itself or an adjacent site. A word of length at most K , starting at the boundary, cannot involve a coefficient below depth K . Thus the boundary entries of the first $K + 1$ powers agree. Subtract the two exponential series and use $\sum_{r \geq K+1} x^r/r! \leq e^x x^{K+1}/(K+1)!$. Integration yields the first bound and the finite- S portion of the second. Uniform stability bounds the remaining integral by $2C_*e^{-\lambda_*S}/\lambda_*$.

For the finite-coordinate perturbation, the generator difference has operator norm at most $4r$, by the tridiagonal row-sum bound and the damping entry. Duhamel's formula gives a bound $4rte^{\Lambda t}$ on the semigroup difference. Integrating it proves the stated d_h bound. Alternatively stability in both factors gives $4rC_*^2te^{-\lambda_*t}$; its integral is $4rC_*^2/\lambda_*^2$. Finally $\|h_\beta - h_\gamma\|_{C([0,T])} \leq \|g_\beta - g_\gamma\|_1$. For a general pair, insert the admissible hybrid with the first parameter's tail and the second parameter's finite coordinates. This proves both locality-plus-perturbation estimates without a factor growing with K . $\square$

For d_h , let $K_h(\epsilon)$ be the least nonnegative integer such that the first right side of (74), multiplied by a , is at most $\epsilon/4$. For d_* , choose

$$S(\epsilon) = \max \left\{ 1, T, \lambda_*^{-1} \log \frac{16(U_* + a)C_*}{\lambda_*\epsilon} \right\}$$

and let $K_*(\epsilon)$ be the least nonnegative integer making

$$2(U_* + a)S(\epsilon)e^{\Lambda S(\epsilon)} \frac{(\Lambda S(\epsilon))^{K+1}}{(K+1)!} \leq \epsilon/8.$$

The remaining stability tail is at most $\epsilon/8$. For either metric d , use its corresponding $K = K_d(\epsilon)$ and $C_d \in \{C_h, C_*^{\text{met}}\}$, and set

$$r_d(\epsilon) = \min\{w_-, 1, \epsilon/(4C_d)\}, \quad \overline{\mathcal{H}}_d(\epsilon) = (2K+3) \log(2+W/r_d(\epsilon)), \quad \mathcal{P}_d(\epsilon) = (2K+3) \log \frac{1}{p-r_d(\epsilon)}. \quad (75)$$

Then $\mathcal{H}_d(\epsilon) \leq \overline{\mathcal{H}}_d(\epsilon)$, and every d -ball of radius ϵ has prior mass at least $e^{-\mathcal{P}_d(\epsilon)}$. Indeed a coordinate mesh of radius $r_d(\epsilon)$, with one fixed tail, covers the response image; each matching coordinate interval has length at least $r_d(\epsilon)$ even at a boundary truth. Independence of the coordinates gives the mass bound, while the unrestricted tail has mass one. The resulting metric error is at most $\epsilon/2$, so the bound holds for open balls as well. In particular $\mathcal{P}_d$ is an upper bound on minus log prior mass.

These explicit formulas imply, with box-dependent constants,

$$\begin{aligned} \mathcal{H}_{d_h}(\epsilon) + \mathcal{P}_{d_h}(\epsilon) &\leq C \frac{(1 + |\log \epsilon|)^2}{\log(2 + |\log \epsilon|)}, \\ \mathcal{H}_{d_*}(\epsilon) + \mathcal{P}_{d_*}(\epsilon) &\leq C(1 + |\log \epsilon|)^2 \quad (0 < \epsilon < 1/2). \end{aligned} \quad (76)$$

For the first assertion use $m! \geq (m/e)^m$ with fixed ΛT : $m = C|\log \epsilon|/\log(2 + |\log \epsilon|)$, for sufficiently large fixed C , makes the factorial tail smaller than $\epsilon/4$. For the second, $S(\epsilon) = O(1 + |\log \epsilon|)$. Taking

$$m \geq \max \left\{ 2e\Lambda S, \frac{\log(16(U_* + a)S/\epsilon) + \Lambda S}{\log 2}, 1 \right\}$$

makes the required tail at most $\epsilon/8$, because $(\Lambda S)^m/m! \leq 2^{-m}$. Thus $K_* = O(1 + |\log \epsilon|)$. In both cases $\log(1/r_d(\epsilon)) = O(1 + |\log \epsilon|)$, proving (76); finitely many larger radii are absorbed in the constant. The finite-radius bounds remain the explicit formulas (75).

Theorem 11.2 (Growing-depth recovery under washed or no-washout policies). *Let $J_n \rightarrow \infty$, $0 < \delta_n \leq 1$, and $q_n = \rho_n a^2 \underline{\kappa}_{J_n}(\delta_n)$, where $\rho_n \in (0, 1]$ is the exploration floor. Use $d = d_h$ for*

the washed protocol, including adaptive nonexploration durations, and $d = d_*$ for the no-washout protocol. If

$$\Psi_n(q_n/16) \rightarrow 0, \quad nq_n \rightarrow \infty, \quad \mathcal{P}_d(\sqrt{q_n}/4) = o(nq_n), \quad (77)$$

then, uniformly over the corresponding horizon-dependent policy class,

$$\Pi_n^{\nu_n} \{d_{J_n}(\beta, \beta_0) \geq \delta_n\} \rightarrow 0 \quad \text{in probability.}$$

A directly checkable sufficient replacement for the first condition is $\mathcal{H}_d(e(q_n/16)) = o(nq_n^2)$ and $nq_n^2 \rightarrow \infty$. For $\inf_n \rho_n > 0$, fixed $0 < \eta < 1$, and

$$(J_n + 1)^3 = o(\log n), \quad \delta_n = \exp\{-\eta \log n / [80(J_n + 1)]\}, \quad (78)$$

all the conditions hold and $\delta_n \rightarrow 0$. This includes the earlier $J_n = o((\log n)^{1/5})$ regime and also permits $J_n = o((\log n)^{1/3})$.

Proof. The squared conditional response contrast on the separated cylinder is at least q_n , by the exploration identity and the inverse theorem. The hypotheses of Theorem 6.3 hold with the deterministic metric just specified. Its failure probability tends to zero under (77), including the transient term by (35); its posterior upper bound tends to zero by the prior-mass condition. This proves the assertion without a fixed-resolution-to-shrinking-resolution interchange.

For the concrete choice, put $j_n = J_n + 1$. The new inverse bound gives

$$q_n \geq (\inf_m \rho_m) a^2 e^{-A_0 j_n^3} n^{-\eta/4} = n^{-\eta/4 - o(1)}.$$

Thus $nq_n^2 \geq n^{1-\eta/2-o(1)}$, whereas both entropy and negative log prior mass at the required radii are $O((\log n)^2)$. These verify the sufficient conditions with all fixed likelihood constants retained. Also $\log n/j_n \rightarrow \infty$, so $\delta_n \rightarrow 0$. All estimates are uniform in the policy and true coefficient sequence. $\square$

Corollary 11.3 (Exponentially weighted operator-norm contraction). *Let $W_\tau e_j = e^{-\tau j} e_j$, $\tau > 0$. With $C_{\text{box}} = 2b_+ + 4a_+$,*

$$\|W_\tau(J_\beta - J_{\beta_0})W_\tau\|_{\text{op}} \leq 3d_J(\beta, \beta_0) + C_{\text{box}} e^{-2\tau(J+1)}. \quad (79)$$

Consequently the preceding theorem with $\delta_n \rightarrow 0$ gives contraction at radius $3\delta_n + C_{\text{box}} e^{-2\tau(J_n+1)}$.

Proof. Split the tridiagonal difference into the entries involving the known coefficients $b_0, \dots, b_J, a_0, \dots, a_J$, including the edge from J to $J+1$, and the remaining entries. The first part has row-sum norm at most $3d_J$, also after multiplication by W_τ . Every index in the second part is at least $J+1$, so its weighted row sum is at most $C_{\text{box}} e^{-2\tau(J+1)}$. The Schur bound proves the inequality. This is explicitly a weighted operator topology, not an assertion of ordinary operator-norm recovery. $\square$

11.2. Anytime frequentist confidence sets. Let $p_q > 0$ be weights on the countable labeled diagnostic cells, summing to one; one may use $p_q = \omega_q$. Fix $0 < \alpha < 1$. For the washed protocol count all visits to cell q ; for the no-washout protocol count exploration visits only. Write $m = N_q(n)$ and $\hat{h}_q = (am)^{-1} \sum_{i \in q, i \leq n} S_i Y_i$ when $m > 0$. Set $\ell_{q,m} = \log\{2m(m+1)/(\alpha p_q)\}$. Define radii

$$R_{q,n}^{\text{wash}} = \frac{\sigma}{a} \sqrt{2\ell_{q,m}/m} + \frac{1}{am} \sum_{i \in q, i \leq n} r_i, \\ R_{q,n}^{\text{lin}} = \frac{\sqrt{2(\sigma^2 + B_*^2)\ell_{q,m}}}{a\sqrt{m}}, \quad B_* = C_*\{R + U_*/\lambda_*\}, \quad (80)$$

with radius $+\infty$ and no constraint when $m = 0$. Here r_i is the undifferentiated washout envelope, not a derivative envelope.

Theorem 11.4 (Simultaneous honest response and coefficient cylinders). *For either protocol and any finite collection $\mathcal{Q}_n$ of cells, let*

$$\mathcal{C}_n = \{\beta \in \mathfrak{B} : |h_\beta(t_q) - \hat{h}_q| \leq R_{q,n} \ (q \in \mathcal{Q}_n)\}.$$

Then $P_{\beta_0, z_0} \{\beta_0 \in \mathcal{C}_n \text{ for all } n\} \geq 1 - \alpha$ for a fixed infinite admissible policy. At each horizon the bound is uniform over horizon-dependent policies. It is also valid for data-selected finite cell collections, since the underlying event controls all cells and times. These are frequentist confidence sets, not posterior credible sets.

For any $J \geq 0$, $0 < \delta \leq 1$, suppose $\mathcal{Q}_n$ includes exactly the required R_J positive-time cells $\{(J, s_J(\delta), k) : 1 \leq k \leq R_J\}$, with any additional cells allowed. Put $\varepsilon = \varepsilon_{J, s_J(\delta)}$. If

$$2 \max_{1 \leq k \leq R_J} R_{(J, s_J(\delta), k), n} < \frac{\delta \varepsilon^{R_J}}{2C_{R_J} L_J}, \quad (81)$$

then $\text{diam}_{d_J} \mathcal{C}_n \leq \delta$. The unmeasured origin is supplied by $h_\beta(0) - h_\gamma(0) = 0$. An empty confidence set has diameter zero; emptiness is excluded on the coverage event.

Proof. For a selected cell, its visit indicator is known before the fresh sign and readout. In the washed experiment the sum of selected $S_i \xi_i$ has conditional moment generating function bounded by the Gaussian one; the remaining signed residual is deterministically bounded by the sum of r_i 's. In the no-washout experiment, at a visit,

$$Y_i = a S_i h_{\beta_0}(t_q) + B_i^0 + \xi_i, \quad |B_i^0| \leq B_*.$$

The intercept is measurable before S_i , so

$$E[e^{\lambda S_i (B_i^0 + \xi_i)} \mid \text{pre-sign information}] = e^{\sigma^2 \lambda^2 / 2} \cosh(\lambda B_i^0) \leq e^{(\sigma^2 + B_*^2) \lambda^2 / 2}.$$

Here Z_n denotes the sum of the selected centered increments just described. In either case $\exp\{\lambda Z_n - \lambda^2 v N_q(n)/2\}$ is a nonnegative supermartingale, where $v = \sigma^2$ or $\sigma^2 + B_*^2$. Stop it at the m th visit, first intersecting with a deterministic time bound and then letting that bound increase. Markov's inequality and optimization in λ show that a two-sided error exceeding $\sqrt{2vm\ell_{q,m}}$ at that visit has probability at most $2e^{-\ell_{q,m}} = \alpha p_q / [m(m+1)]$. This argument does not assert that a sum conditioned on an adaptively chosen visit count is Gaussian. Summing over all $m \geq 1$ and all cells gives total failure probability at most α . Division by am , and the washed residual budget, prove simultaneous coverage.

For two elements of $\mathcal{C}_n$, the response difference on every required positive grid point is at most twice its radius. The zero point has difference zero. The finite-grid certificate and scale choice bound all derivatives by a value strictly less than δ/L_J . The all-depth inverse then gives $d_J < \delta$, proving the diameter bound. Every threshold uses the same corrected L_J . $\square$

For each fixed grid, exploration-only counts divided by n converge to $\rho\omega_q > 0$ under a fixed positive floor. Therefore the visit counts in either protocol satisfy $\liminf_n N_q(n)/n \geq \rho\omega_q$, and the radii tend to zero, and the fixed-block diameter condition eventually holds. The coverage event remains the single event of probability at least $1 - \alpha$; it is not obtained by intersecting fixed-time coverage claims. For growing grids the explicit threshold remains a sufficient condition. In the no-washout experiment the entire construction costs at most $\tau_+ n$ physical time.

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